

Document Title : **Asset Life Extension Study for Grissik Plant - Assessment Report for Static Equipment**

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REVISION SHEET

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TABLE OF CONTENTS

REVISION SHEET	1
TABLE OF CONTENTS	2
LIST OF FIGURES	3
LIST OF TABLES.....	4
1 EXECUTIVE SUMMARY.....	5
2 INTRODUCTION.....	9
2.1 Background	9
2.2 Objective	9
2.3 Scope of Works	10
2.4 Limitations	10
2.5 Abbreviations.....	11
3 TECHNICAL APPROACH AND METHODOLOGY.....	12
3.1 Governing Criticality for Each Equipment.....	13
4 KEY GENERIC DATA FOR ASSESSMENT AND ASSUMPTIONS.....	14
5 DAMAGE MECHANISM REVIEW	15
5.1 Internal Corrosion	15
5.2 External Corrosion.....	15
5.3 Environment Assisted Cracking	16
5.4 Mechanical & Metallurgical Failure	16
5.5 Minimum Required Thickness.....	17
5.6 Corrosion Rate	18
5.7 Remaining Useful Life.....	18
6 ASSESSMENT RESULTS	19
6.1 IAAN List for Static Equipment.....	19
6.2 Decision Matrix Criticality & RBI Risk Summary.....	19
6.3 Static Equipment Component Directly Impacted by Post Tie-in Summary	20
6.4 Static Equipment Indirectly Impacted by Post Tie-in Summary	22
6.5 Static Equipment Component Not Impacted by Post Tie-in Summary.....	24
6.6 RUL Static Equipment Not Impacted by Post Tie-in Summary.....	24
7 CONCLUSIONS AND RECOMMENDATIONS FOR STATIC EQUIPMENT	26
7.1 Conclusions	26
7.2 Recommendations.....	27
APPENDIX A – STATIC EQUIPMENT GENERAL DATA.....	28
APPENDIX B – LATEST RBI ASSESSMENT RESULTS.....	29
APPENDIX C – REMAINING USEFUL LIFE ASSESSMENT RESULT	30
APPENDIX D - LIFE EXTENSION PROGRAM BASED ON RISK AND EQUIPMENT CRITICALITY	31
APPENDIX E – IAAN LIST.....	32

LIST OF FIGURES

Figure 1-1 Static equipment 2027 ALE priority on component level.....	8
Figure 2-1 Block flow diagram of Grissik plant after post tie-in	9
Figure 6-1 RBI Risk static component direct impacted by post tie-in summary	20
Figure 6-2 Static component directly impacted by post tie-in 2027 ALE Priority	21
Figure 6-3 RBI Risk static component indirectly impacted by post tie-in summary	22
Figure 6-4 Static equipment indirectly impacted by post tie-in ALE priority	23
Figure 6-5 RBI Risk static component not impacted by post tie-in summary	24
Figure 6-6 Static component not impacted by post tie-in 2027 ALE Priority.....	25

LIST OF TABLES

Table 1-1 Grissik plant static component criticality (directly impacted by post tie-in)	6
Table 1-2 Grissik plant static component criticality (indirectly impacted by post tie-in).....	6
Table 1-3 Grissik plant static component criticality (not impacted by post tie-in).....	6
Table 1-4 Component list with criticality 2	6
Table 1-5 Component list with criticality 3	7
Table 2-1 Scope of work static equipment (equipment and component count).....	10
Table 2-2 Scope of work static component affected by post tie-in	10
Table 2-3 Components excluded	10
Table 3-1 ALE Priority	12
Table 3-2 Priority definition.....	12
Table 6-1 IAAN List for static equipment	19
Table 6-2 Decision matrix in Criticality & RBI Risk	20
Table 6-3 RUL static component directly impacted by post tie-in summary	20
Table 6-4 RUL static equipment directly impacted by post tie-in summary	21
Table 6-5 Decision matrix for static component directly impacted by post tie-in	21
Table 6-6 RUL static component indirectly impacted by post tie-in summary	22
Table 6-7 RUL static equipment indirectly impacted by post tie-in summary	22
Table 6-8 Decision matrix for static component indirectly impacted by post tie-in.....	23
Table 6-9 RUL static component not impacted by post tie-in summary	24
Table 6-10 RUL static equipment not impacted by post tie-in summary	24
Table 6-11 Decision matrix for static component not impacted by post tie-in.....	25

1 EXECUTIVE SUMMARY

The design and operating conditions of 132 static equipment items comprising 175 assessed pressure-retaining components were evaluated as part of this assessment. The evaluation covered the following activities:

1. Reviewing the condition of static equipment, including pressure vessels, air coolers, columns, heat exchangers, filters, pig launchers, and receivers, based on a comprehensive assessment of original design information, design changes, operating conditions, future service requirements, and inspection history;
2. Prioritizing and ranking assets based on probability of failure, damage mechanisms, and consequence of failure, using the Risk-Based Inspection assessment results performed by MEPG;
3. Identifying static equipment and components with progressive damage mechanisms and evaluating their remaining useful life (RUL); and
4. Recommending actions to improve the probability of safe and uninterrupted equipment operation, including inspection and maintenance practices that support the life extension program.

Based on these activities, the ALE assessment determined the criticality of each component of equipment and proposed specific inspection and maintenance actions to support continued operation and life extension through 2040 under the post tie-in condition. This assessment was performed with limitations and assumptions, as described in section 2.4. A site survey was not conducted for this study to verify the current equipment condition; therefore, MEPG's latest inspection reports and PCMS data were used as the basis for assessing the general condition of the equipment.

Risk Based Inspection (RBI) Assessment Result Review

The RBI assessment result data has been reviewed and can be summarized as follows:

1. MEPG has implemented an extensive RBI program for the CGP Plant. Based on the document review and discussions with MEPG's Asset Integrity team, the RBI program follows the methodology outlined in API 580 and API 581. It is recommended that MEPG continue this program to determine active damage mechanisms, identify risk levels, and establish suitable inspection programs and inspection intervals through 2040.
2. Based on the projected risk profiles for 2033 and 2040, several static equipment items are expected to show an increase in risk level, including changes from Low to Medium I, Medium I to Medium II, and Medium II to High. These projections are based on the RBI risk condition if inspection is not conducted. It is recommended that MEPG improve the RBI report by incorporating the projected future risk. This is important to evaluate the effect of the recommended inspections on the overall risk level. By adopting this recommendation, MEPG will be able to determine the inspection effectiveness required for each equipment item to bring risk within the tolerable range during the life extension period, including operation under post tie-in conditions.
3. The latest findings from all the inspection performed in MEPG should be incorporated and integrated with the RBI assessment including external inspection, internal inspection, coating inspection and CUI inspection. The findings from these inspections will give higher confidence in the assessment and overall risk result.

RUL Assessment and Life Extension Program

ALE program recommending the necessary actions based on the respective criticality of the equipment was developed. The RUL criticality criteria were defined by considering the RUL against the extension life target as shown in Table 1-1 through Table 1-3.

Table 1-1 Grissik plant static component criticality (directly impacted by post tie-in)

Criticality	Remaining Useful Life	Number of Component	Percentage of Component
Criticality 1	RUL 2026 $\leq$ 0 yr	0	0
Criticality 2	0 < RUL 2026 $\leq$ 6 yr (to 2032)	0	0
Criticality 3	6 < RUL 2026 $\leq$ 14 yr (to 2040)	2	7.4
Criticality 4	RUL 2026 > 14 yr	25	92.6
Total		27	100

Table 1-2 Grissik plant static component criticality (indirectly impacted by post tie-in)

Criticality	Remaining Useful Life	Number of Component	Percentage of Component
Criticality 1	RUL 2026 $\leq$ 0 yr	0	0
Criticality 2	0 < RUL 2026 $\leq$ 6 yr (to 2032)	1	0.89
Criticality 3	6 < RUL 2026 $\leq$ 14 yr (to 2040)	14	12.5
Criticality 4	RUL 2026 > 14 yr	97	86.61
Total		112	100

Table 1-3 Grissik plant static component criticality (not impacted by post tie-in)

Criticality	Remaining Useful Life	Number of Component	Percentage of Component
Criticality 1	RUL 2026 $\leq$ 0 yr	0	0
Criticality 2	0 < RUL 2026 $\leq$ 6 yr (to 2032)	0	0
Criticality 3	6 < RUL 2026 $\leq$ 14 yr (to 2040)	0	0
Criticality 4	RUL 2026 > 14 yr	36	100
Total		36	100

Inspection and monitoring plans were developed to define the required inspection intervals for each equipment item based on its respective RUL criticality. The inspection and monitoring requirements for each criticality level are summarized below:

1. Criticality 1
No static equipment or components at the Grissik Plant are classified as Criticality 1.
2. Criticality 2
One component is classified as Criticality 2 with indirectly impacted by post tie-in condition, listed in Table 1-4. The governing factor is its corrosion rate of 0.65 mm/year. It is strongly recommended that MEPG verify the component thickness before determining any further action.

Table 1-4 Component list with criticality 2

Equipment ID	Insulated (Y/N)	Equipment Type	Affected Category	Component Criticality
35-V-101 SHELL	Y	Vessel	Indirectly Impacted	Criticality 2

3. Criticality 3
16 components are classified as Criticality 3, with 2 components directly impacted by post tie-in condition as listed in the Table 1-5 below. The primary contributing factor for this classification is the high corrosion rate identified for these components. It is recommended that MEPG perform inspections in accordance with the existing RBI plan to verify the current wall thickness.

Table 1-5 Component list with criticality 3

Equipment ID	Insulated (Y/N)	Equipment Type	Affected Category	Component Criticality
20-V-17.03	N	Filter	Indirectly Impacted	Criticality 3
21-E-202 SHELL	Y	Exchanger	Indirectly Impacted	Criticality 3
24-MAF-401 BOTTOM	N	Column	Directly Impacted	Criticality 3
25-E-202A SHELL	Y	Exchanger	Indirectly Impacted	Criticality 3
25-V-202 BOTTOM	Y	Vessel	Indirectly Impacted	Criticality 3
25-V-202 TOP	Y	Vessel	Indirectly Impacted	Criticality 3
25-V-203	N	Vessel	Directly Impacted	Criticality 3
25-V-205D BOTTOM	Y	Vessel	Indirectly Impacted	Criticality 3
25-V-205D TOP	Y	Vessel	Indirectly Impacted	Criticality 3
27-HBG-401 SHELL	N	Exchanger	Indirectly Impacted	Criticality 3
41-S-201 B	N	Filter	Indirectly Impacted	Criticality 3
41-S-302	N	Filter	Indirectly Impacted	Criticality 3
47-MAJ-401-B	N	Filter	Indirectly Impacted	Criticality 3
48-V-101 HEAD A	N	Vessel	Indirectly Impacted	Criticality 3
48-V-101 HEAD B	N	Vessel	Indirectly Impacted	Criticality 3
48-V-101 SHELL	N	Vessel	Indirectly Impacted	Criticality 3

4. Criticality 4

A total of 158 static equipment component is classified as Criticality 4. Of these, 34 components are categorized as Low risk, 37 components categorized as Medium I, 71 components categorized as Medium II, and 16 components categorized as High. All 158 static components are considered suitable for operation until 2040 with continued monitoring.

ALE Priority

Static Equipment 2027 ALE priority presents distribution of static equipment on component level with general conditions (directly impacted, indirectly impacted and not impacted by post tie-in) across priority 1 (P1) to priority 5 (P5). Based on the combination of remaining useful life (RUL) and risk-based inspection (RBI). Figure 1-1 provides a visual representation of the inspection and integrity management priorities required to support Asset Life Extension (ALE) program through 2040.

1. Priority 1

No static equipment components are classified as Priority 1. This indicates that, based on the ALE assessment, no components currently require immediate repair, replacement, or Fitness-for-Service assessment.

2. Priority 2

A total of 21 components is classified as Priority 2, comprising by 2 components directly impacted, 16 components indirectly impacted and 3 components not impacted by the post tie-in condition. These components should be inspected before end of 2027 to confirm their continued Fitness-for-Service and to determine whether any mitigation measures are required.

3. Priority 3

A total of 83 components is classified as Priority 3, comprising by 9 components directly impacted, 55 components indirectly impacted and 19 components not impacted by the post tie-in condition. Although these components have sufficient remaining life for continued operation, periodic monitoring and verification are required to confirm corrosion rate assumptions and integrity performance throughout the life-extension period.

4. Priority 4

A total of 37 components is classified as Priority 4, comprising by 9 component directly impacted, 21 components indirectly impacted and 7 components not impacted by the post tie-in condition. These components are considered suitable for continued operation under the existing RBI program and can be managed through normal inspection intervals without additional intervention.

5. Priority 5

A total of 34 components is classified as Priority 5, comprising by 7 components directly impacted, 20 components indirectly impacted and 7 components not impacted by the post tie-in condition. These components are expected to remain fit for service throughout the ALE period and therefore only require routine monitoring in accordance with the established RBI program.

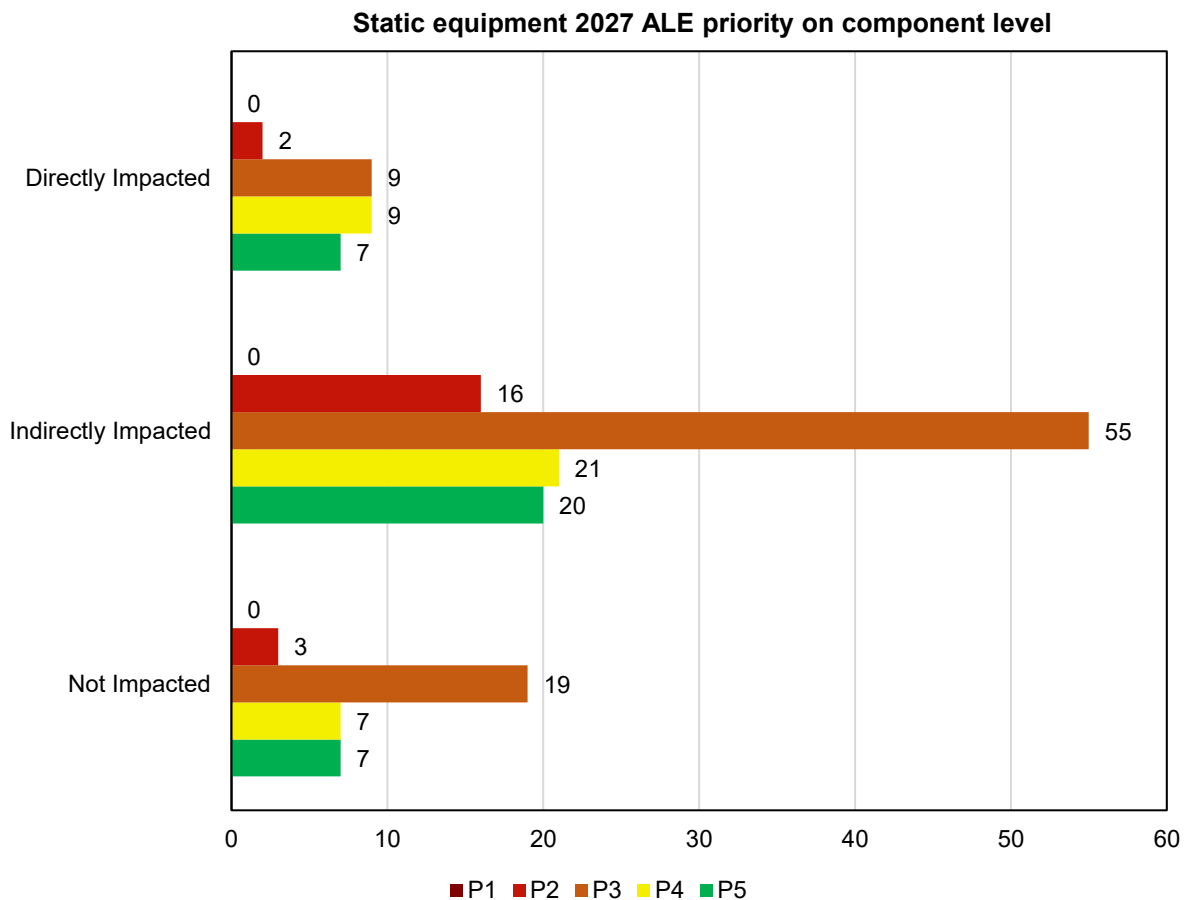


Figure 1-1 Static equipment 2027 ALE priority on component level

2 INTRODUCTION

2.1 Background

The Grissik Plant has been in continuous operation since 1998, with an original nominal design life of 20 years (to 2018). In 2017, an Asset Life Extension (ALE) study conducted by Det Norske Veritas (DNV GL) on behalf of ConocoPhillips Grissik Ltd. confirmed that the plant could be safely extended to 2028, subject to specified inspection and monitoring actions. The facility has since been operated by Medco E&P Grissik (MEPG).

To support the post tie-in operating phase as shown in Figure 2-1, the objective of this 2026 ALE study is to evaluate the next extension target of 2040. This assessment specifically focuses on determining the Remaining Useful Life (RUL) of the major pressure retaining components of static equipment, utilizing historical thickness and corrosion rate data from Plant Condition Management Software (PCMS). As this study is just one component of MEPG's broader asset-integrity framework, it complements and does not replace dedicated inspection and maintenance schemes for specific items such as valves, pressure safety valves (PSVs), flange bolting, gaskets, structural elements, and instrumentation.

The technical analysis utilizes ASME Section VIII Div 1 (vessels) for thickness minimum calculation and projects the RUL to 2040 using corrosion rates derived from measured thickness histories. To ensure direct comparability with historical baselines, this 2026 assessment largely adopts the structure, core methodology, and reporting conventions of the 2017 ALE. However, the current study also incorporates supplementary methodologies to expand the analytical scope and address updated operational requirements.

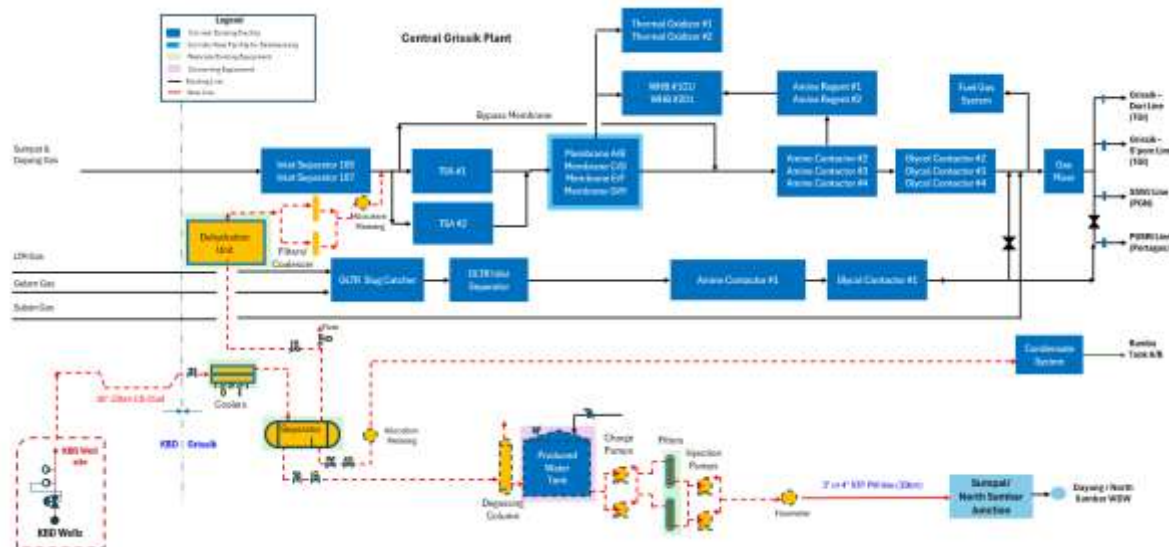


Figure 2-1 Block flow diagram of Grissik plant after post tie-in

2.2 Objective

The primary objective of this study is to evaluate the long-term integrity and remaining service capability of the Grissik Plant facilities and to develop a structured ALE roadmap. This roadmap will include recommended actions, inspection and monitoring strategies, and mitigation measures required to support continued safety, reliability, and compliant operation of the plant through the proposed extended service period until 2040. The assessment objectives are as follows:

1. To evaluate the condition of static equipment, including pressure vessels, air coolers, columns, heat exchangers, filters, launchers, and receivers, through a comprehensive review of original design data, current and historical operating conditions, future service requirements, inspection history and the impact of post tie-in conditions;
2. To prioritize and rank the assets based on their probability of failure, damage mechanisms, consequence of failure, and risk level, using the Risk-Based Inspection assessment results performed by MEPG;

3. To identify static equipment and components with progressive damage mechanisms and evaluate their RUL; and
4. To recommend the actions required to support safe continued operation inspection, fitness-for-service assessment, repair or replacement where required, and monitoring including any that fall outside the existing RBI interval.

2.3 Scope of Works

The scope of works in this assessment is described as follows:

1. Data and document review
Perform review relevant Grissik documentation, including design and operating documents, inspection reports, anomaly reports, and other supporting records.
2. Remaining Useful Life (RUL) assessment
Perform RUL calculations for the static equipment listed in Table 2-1. The scope of static equipment affected by the post tie-in condition is presented in Table 2-2.

Table 2-1 Scope of work static equipment (equipment and component count)

Equipment Type	Component		Equipment	
	Not Insulated	Insulated	Not Insulated	Insulated
Air Cooler	5	0	5	0
Column	16	5	9	3
Exchanger	9	16	9	15
Filter	33	0	30	0
Launcher/Receiver	2	0	2	0
Vessel	56	33	39	20
Sub Total	121	54	94	38
Total	175		132	

Table 2-2 Scope of work static component affected by post tie-in

Equipment Type	Directly Impacted	Indirectly Impacted	Not Impacted
Air Cooler	0	4	1
Column	10	8	3
Exchanger	1	21	3
Filter	4	24	5
Launcher/Receiver	0	0	2
Vessel	12	55	22
Sub Total	27	112	36
Total	175		

3. Life extension program
Based on the calculated remaining life, criticality, risk ranking, previous inspection history, and post tie-in impact conditions, the ALE assessment will recommend future actions required to support life extension of these items to 2040.
4. Review of RBI assessment results
Review the most recent and future RBI assessment results performed by MEPG.
5. Criticality
To determine the criticality level of related equipment based on RUL assessment results.

2.4 Limitations

Components excluded in this assessment are shown in Table 2-3:

Table 2-3 Components excluded

Component	Follow up recommendation
Flange bolting & gaskets	ASME PCC-1 bolting + gasket program
Valve internals & PSV	Valve overhaul program
Heat tracing & instrumentation	Electrical/instrumentation program
Structural support & saddles	Civil/structural inspection
Buried pipelines & rigid risers	Pipeline ILI study (Suban PL)

2.5 Abbreviations

ALE	Asset Life Extension
API	American Petroleum Institute
ASME	American Society of Mechanical Engineers
BPVC	Boiler and Pressure Vessel Code (ASME BPVC)
CA	Corrosion Allowance
CGP	Central Gas Plant
CML	Corrosion Monitoring Location
CR	Corrosion Rate
CUI	Corrosion Under Insulation
DNV GL	Det Norske Veritas GL
FFS	Fitness-for-Service
IAAN	Integrity Anomaly Alert Notification
IDMS	Integrity Data Management System
MEPG	Medco E&P Grissik
N/A	Not Applicable
PAUT	Phased Array Ultrasonic Testing
PCMS	Plant / Process Condition Monitoring System
PSV	Pressure Safety Valve
RBI	Risk Based Inspection
RUL	Remaining Useful Life
TSA	Thermal Swing Adsorber
UT	Ultrasonic Testing

3 TECHNICAL APPROACH AND METHODOLOGY

The 2026 ALE priority is a designed to prioritize asset integrity activities by combining RUL and RBI to determine recommendations activity as shown in Table 3-1 and Table 3-2.

1. Priority 1
Priority 1 represents the highest urgency level and applies to equipment with critical remaining useful life and/or elevated RBI risk. This priority includes all Criticality 1 equipment and Criticality 2 equipment with Medium II or High risk.
2. Priority 2
Priority 2 applies to equipment with limited remaining useful life. This priority includes Criticality 2 equipment with Low or Medium I risk, as well as Criticality 3 and Criticality 4 equipment with High risk.
3. Priority 3
Priority 3 applies to equipment with moderate criticality and acceptable remaining life. This priority includes Criticality 3 equipment with Low, Medium I, or Medium II risk, and Criticality 4 equipment with Medium II risk.
4. Priority 4
Priority 4 applies to equipment with long remaining useful life and moderate RBI risk, specifically Criticality 4 equipment with Medium I risk.
5. Priority 5
Priority 5 applies to equipment with remaining useful life greater than 14 years and Low RBI risk. These items are considered suitable for continued monitoring under the established RBI program.

Table 3-1 ALE Priority

Risk Criticality	Low	Medium I	Medium II	High
Criticality 1 (RUL < 0 yrs)	P1	P1	P1	P1
Criticality 2 (0 < RUL ≤ 6 yrs)	P2	P2	P1	P1
Criticality 3 (6 < RUL ≤ 14 yrs)	P3	P3	P3	P2
Criticality 4 (RUL > 14 yrs)	P5	P4	P3	P2

Table 3-2 Priority definition

Priority	Criticality / Risk	Recommended Activity	Inspection Date
P1	C1 any C2 MEDIUM II C2 HIGH	Immediate inspection, then fitness-for-service, repair or replacement as the result requires. Do not wait for an outage window if the result demands otherwise.	Before end 2026
P2	C2 LOW / MEDIUM I C3 HIGH C4 HIGH	Urgent Inspection before end of 2027	Before end 2027
P3	C3 MEDIUM I C3 MEDIUM II C4 MEDIUM II C3 LOW	Inspect at half the remaining useful life from the assessment date, or at the RBI due date, whichever is earlier	Before end 2033
P4	C4 MEDIUM I	Follow the routine RBI programme at the scheduled date	Per RBI date
P5	C4 LOW	Monitor only, no action beyond the standing RBI interval.	Per RBI date

Note:

The RUL is calculated from the last inspection date and reported against the 2026 baseline, while criticality and priority are assessed at the 2027 post tie-in datum.

3.1 Governing Criticality for Each Equipment

As the criticality rating is based on component level, the equipment's governing criticality rating is chosen as follows:

1. Most severe criticality governs - if multiple components within the same static equipment have different criticality ratings, the most severe criticality level shall govern the equipment rating. For example, if one component is classified as Criticality 4 and another component as Criticality 2, the equipment shall be assigned Criticality 2.
2. If components have the same criticality rating, the governing minimum required thickness (t_{min}) will be used for the RUL assessment. Furthermore, the corrosion rate shall be selected in the following order of priority:
 - a. Measured corrosion rate.
 - b. Plant Condition Management Software (PCMS) corrosion rate.
 - c. Expert Corrosion Rate.
3. For static equipment with the same RUL value, the governing component shall be selected based on the main pressure-retaining component, in the following order of priority:
 - a. Shell.
 - b. Head.

4 KEY GENERIC DATA FOR ASSESSMENT AND ASSUMPTIONS

List of generic data for assessment and assumptions to be used in this assessment listed as follows:

1. Grissik Plant was commissioned in 1998, and the end of design life is in 2018.
2. The next extension life target is 2040.
3. Based on Decision of the Ministry of Energy and Mineral Resources Republic of Indonesia No.: 183.K/HK.02/DMT/2024, the maximum extension of service life is according to the design service life limit. In this case the design life of Grissik plant is 20 years, therefore for equipment which is have RUL more than 20 years, will be noted as ">20 years".
4. Limited inspection reports are available to assess the equipment. Therefore, the assessment has been performed with limitations and is based on certain assumptions as described in this section.
5. Site survey has not been performed for this study to evaluate the current external condition of the equipment / lagging / fabric condition. The MEPP Grissik Plant latest inspection reports are used as the basis to assess the general condition.
6. The focus of this study was to identify RUL, Criticality and ALE priority. However, it does not consider all possible operational issues and scenarios outside of the safe operating limits.
7. The study does not consider operational scenarios outside the safe operating limits.
8. All the data extracted from PCMS is assumed to be in high accuracy.
9. The corrosion rates used for the RUL calculations were considered into 3 hierarchy as listed below:
 - a. Measured corrosion rate.
 - b. PCMS corrosion rate.
 - c. Expert corrosion rate.
10. The corrosion rate used in the assessment is considered to be constant throughout the study period (2027 to 2040).
11. Remaining life is not calculated based on other damage mechanisms such as mechanical & metallurgical failure and any environmental assisted cracking. Findings related to these from document review are recorded in the life extension program for further assessment in the next phase.
12. Minimum required thickness (t_{min}) for each vessel/pipe is based on data from PCMS. In the absence of this information, (t_{min}) is calculated per ASME Section VIII Div 1 (vessels).
13. This assessment only considers the risk (current and future) and the remaining life based on internal and external wall loss (corrosion).
14. The remaining life for the equipment is taken as the lowest remaining life among the components.
15. The post tie-in impact on equipment is classified into the following three categories:
 - a. Directly Impacted
The tie-in changes the fluid service in these components.
 - b. Indirectly Impacted
The process of fluid within this condition does not mix with the post tie-in fluid, failure of the equipment has the potential to affect Sakakemang operations and production continuity.
 - c. Not Impacted
No change in service arising from the tie-in.

5 DAMAGE MECHANISM REVIEW

5.1 Internal Corrosion

The 2017 ALE found internal thinning to be the dominant damage mechanism 62.8% of Grissik component damage mechanisms were governed by internal/external loss of thickness, primarily CO₂ corrosion. The ALE 2026 assessment confirms the same mode. Type of internal corrosion damage mechanism involved in an ALE 2026 assessment for static equipment is:

1. Amine Corrosion
This is typically a localized form of corrosion that occurs principally on carbon steel in amine treating processes. The corrosion is not caused by the amine itself but rather results from dissolved acid gases (CO₂ and H₂S), heat stable amine salts (HSAS), amine degradation products (such as bicine, oxalate, and formate salts), and other process contaminants. It often manifests as uniform thinning in localized areas or under-deposit attack, especially in areas of high velocity or turbulence.
2. CO₂ Corrosion
This mechanism results when CO₂ dissolves in water to form carbonic acid (H₂CO₃). The acid lowers the pH of the water, and sufficient quantities can promote highly aggressive general corrosion and/or pitting corrosion of carbon steel and low-alloy steels. It often occurs in locations where a cooling effect causes condensation of the water and acid from the vapor phase.
3. High/Low Temperature H₂S/H₂ Corrosion
This form of corrosion occurs in piping and equipment where high-temperature H₂/H₂S streams are found, primarily in hydro processing units like hydrotreaters and hydrocrackers. It causes uniform wall thinning of carbon and low-alloy steel. The presence of hydrogen accelerates the sulfidation process compared to environments where only H₂S is present without hydrogen.

The unspecified internal corrosion cannot be explained by the mechanism, equipment or component with this damage mechanism needs to be investigated.

5.2 External Corrosion

External corrosion of static equipment, other than CUI, which is addressed in a dedicated chapter, arises from the interaction of the coating system with the plant environment. Where the coating remains intact the external-corrosion rate is essentially zero; where the coating has broken down, external metal loss can proceed rapidly and localize at any surface where water can persist. The condition of the external coating system is therefore the governing control for external corrosion. The 2017 ALE recommendation to maintain the external coating system and to inspect visually during planned shutdowns is retained without modification. In line with the interim scope, visual inspection findings continue to be logged into the PCMS anomaly workflow, and any coating-deterioration or metal-loss finding classified as significant is progressed through the IAA process. Type of external corrosion damage mechanism involved in an ALE 2026 assessment for static equipment is:

1. CUI Carbon and Low Alloy Steels
CUI is the corrosion of piping, pressure vessels, and structural components resulting from water becoming trapped under insulation or fireproofing. The corrosion rate increases with metal temperature up to the point where the water evaporates quickly. It typically appears as rough, uneven, and somewhat pitted localized thinning, usually covered with a loose, flaky, non-protective scale.
2. External Corrosion Carbon and Low Alloy Steels
Atmospheric corrosion is the deterioration that occurs from moisture associated with atmospheric conditions. It affects unpainted or uninsulated carbon and low-alloy steel equipment operating below 250 °F (120 °C) where moisture can contact the surface. Environments that are marine, moist, or heavily polluted with airborne contaminants (like chlorides or sulfur compounds) cause the most severe atmospheric corrosion.

5.3 Environment Assisted Cracking

Environment-assisted cracking is a degradation mechanism in which an otherwise ductile metal develops brittle cracks due to the combined effect of three required factors: susceptible material, tensile stress, and a specific corrosive environment. Susceptible materials may include carbon steel with high-hardness weld areas or certain grades of stainless steel. Tensile stress may result from operating pressure, external loads, or residual stresses introduced during welding and fabrication. The corrosive environment may include chlorides, wet H₂S, caustic, or amine service. If any one of these three factors is eliminated, cracking will not occur. Environment-assisted cracking is particularly critical because it can develop with little or no general wall thinning, meaning the equipment may appear acceptable until cracking initiates or propagates. Type of environment assisted cracking corrosion damage mechanism involved in an ALE 2026 assessment for static equipment is:

1. Chloride Stress Corrosion Cracking – External
This is surface-initiated cracking of 300 series stainless steels and some nickel-based alloys operating under the combined action of tensile stress, elevated temperature, and an aqueous chloride environment. External Cl⁻ SCC frequently occurs on insulated stainless-steel surfaces when the insulation gets wet and traps chlorides against the hot metal surface. The cracks typically have many branches and are trans granular in nature.
2. Wet H₂S (HIC/SOHIC/Blistering)
Results from hydrogen atoms (generated by the wet H₂S corrosion process) diffusing into the steel and collecting at internal discontinuities like inclusions or laminations. The atoms combine into H₂ gas molecules, building pressure until local deformation forms a bulge or blister, primarily on the ID surface. Hydrogen-Induced Cracking (HIC) & Stress-Oriented HIC (SOHIC): Similar to blistering, but the trapped hydrogen gas pressure causes internal stepwise cracking that links up through the thickness of the steel (HIC) or aligns in arrays perpendicular to the primary tensile stress (SOHIC), potentially causing structural failure.
3. Wet H₂S (Sulfide Stress Cracking)
SSC is a form of hydrogen embrittlement cracking that occurs in high-strength steels or localized hard zones (like non-post-weld-heat-treated weld heat-affected zones) when exposed to a wet H₂S environment. Atomic hydrogen diffuses into the susceptible hard microstructure, embrittling the metal matrix and causing cracks to initiate and propagate rapidly under tensile stress.

5.4 Mechanical & Metallurgical Failure

Unlike environment-assisted cracking, mechanical and metallurgical failures do not require a specific corrosive environment to occur. Mechanical failure is driven by physical forces such as stress, pressure, vibration, or cyclic loading that can deform, crack, or rupture a component. In contrast, metallurgical failure occurs when the metal's internal microstructure changes because of operating conditions, typically prolonged exposure to elevated temperatures. These microstructural changes can degrade the mechanical properties of the material, reduce its strength, toughness, or ductility and make it more susceptible to brittle or unexpected failure. Type of mechanical & metallurgical failure damage mechanism involved in an ALE 2026 assessment for static equipment is:

1. Cladding
While "cladding" is not a standalone damage mechanism in API 571, clad materials (like a stainless-steel layer on a carbon steel backing) can fail mechanically through Dissimilar Metal Weld (DMW) Cracking or Thermal Fatigue. Differential thermal expansion between the cladding and the base metal during thermal cycling can induce cyclical stress, ultimately leading to mechanical or thermal fatigue cracking at the bond line or associated welds.
2. Mechanical Fatigue
Fatigue cracking is a mechanical form of degradation that occurs when a component is exposed to cyclical stresses for an extended period, such as from dynamic loading due

to vibration, water hammer, or unstable fluid flow. It often results in sudden, unexpected through-wall cracking. Cracks generally initiate stress concentration points, such as notches, surface defects, or welds.

5.5 Minimum Required Thickness

Minimum required thickness (t_{min}) per component or equipment is taken directly from the PCMS retiring limit with fallback to the calculation-derived t_{min} . If t_{min} is not available from PCMS data, the 2026 ALE calculated t_{min} per ASME Section VIII Div 1 for static equipment will be used as minimum required thickness.

1. Shell minimum required thickness when the thickness does not exceed one-half the inside radius, or P does not exceed $0.385 SE$ (Circumferential Stress / Longitudinal Joints), is:

$$t_{ms} = \frac{PR}{SE - 0.6P}$$

Where,

P = internal design pressure

R = inside radius of the shell course under consideration

S = maximum allowable stress (ASME BPVC section II, part D)

E = joint efficiency for, or the efficiency of, appropriate joint in cylindrical or spherical shells, or the efficiency of ligaments between opening, whichever is less.

2. Shell minimum required thickness when the thickness does not exceed one-half the inside radius, or P does not exceed $1.25 SE$ (Longitudinal Stress/Circumferential Joints), is:

$$t_{ms} = \frac{PR}{2SE + 0.4P}$$

Where,

P = internal design pressure

R = inside radius of the shell course under consideration

S = maximum allowable stress (ASME BPVC section II, part D)

E = joint efficiency for, or the efficiency of, appropriate joint in cylindrical or spherical shells, or the efficiency of ligaments between opening, whichever is less.

3. Ellipsoidal head, t_{min} is:

$$t_{meh} = \frac{PD}{2SE - 0.2P}$$

Where,

P = internal design pressure

D = inside diameter of the head at skirt

S = maximum allowable stress (ASME BPVC section II, part D)

E = joint efficiency for, or the efficiency of, appropriate joint in cylindrical or spherical shells, or the efficiency of ligaments between opening, whichever is less.

4. Hemispherical head, t_{min} is:

$$t_{mhh} = \frac{PL}{2SE - 0.2P}$$

Where,

P = internal design pressure

L = inside spherical or crown radius

S = maximum allowable stress (ASME BPVC section II, part D)

E = joint efficiency for, or the efficiency of, appropriate joint in cylindrical or spherical shells, or the efficiency of ligaments between opening, whichever is less.

5. Torispherical head, t_{min} is:

$$t_{mth} = \frac{0.885PL}{SE - 0.1P}$$

Where,

P = internal design pressure

L = inside spherical or crown radius

S = maximum allowable stress (ASME BPVC section II, part D)

E = joint efficiency for, or the efficiency of, appropriate joint in cylindrical or spherical shells, or the efficiency of ligaments between opening, whichever is less.

5.6 Corrosion Rate

The internal and external corrosion rates assumed for general and localized thinning are based on available measured data, PCMS data, or corrosion rates calculated as part of the ALE assessment.

1. Measured Corrosion Rate

The measured corrosion rate was determined from available wall-thickness inspection data and corrosion rate information provided by MEPG.

2. PCMS Corrosion Rate

The PCMS corrosion rate was applied where a measured corrosion rate could not be derived from the available inspection data.

3. Expert Corrosion Rate

The expert corrosion rate calculation was based on the mean corrosion rate at the CML, the worst observed metal loss for similar equipment or components, or an adjusted corrosion rate where the PCMS corrosion rate was too low to support meaningful RUL calculation for static equipment. This calculation approach was applied when neither measured corrosion rates nor PCMS corrosion rates were available.

5.7 Remaining Useful Life

Deterministic RUL is computed as follows:

$$RUL = \frac{(t_{actual} - t_{min})}{CR}$$

Where,

t_{actual} = the measured thickness of component in mm, if the measured thickness is not available, $t_{nominal}$ will be used.

t_{min} = refer to section 5.5

CR = refer to section 5.6

6 ASSESSMENT RESULTS

6.1 IAAN List for Static Equipment

IAAN list for static equipment is shown in Table 6-1.

Table 6-1 IAAN List for static equipment

Equipment ID	Equipment Type	Description
21-V-201 D	Vessel	Leakage on 10" outlet nozzle (N2) TSA201D
21-V-202	Vessel	Seepage on 2" condensate line Grissik
25-E-103	Air Cooler	IAAN: INLET LINE OF COOLER 25-E-103C/D LEAK
25-V-202	Vessel	IAAN: Leakage on Weld Joint of Heat Trac
25-V-205 D	Vessel	LEAKAGE ON DS 25-LV-226 STEAM CONDENSATE
35-E-102	Exchanger	Small Fugitive on Flange Connection
25-C-201	Vessel	Flange bolt corroded

Additional information about IAAN from Grissik thermal swing absorber update in February 2025, can be summarized:

- Thermal Swing Absorbers: 21-V-101 A/B/C/D and 21-V-201 A/B/C/D with general information as follows:
 - Material: SA 516-70N carbon steel.
 - Design pressure/ temperature: 1250 psig / 650 °F.
 - Dimension: 30 ft length × 7'-8" diameter.
 - Nominal thickness: 3.54" + CA 0.125".
 - Year built: 1998.
- Failure history:
 - 2011-2012: 21-V-201D failed after 11 yr (repaired). Other vessels inspected clean at the time.
 - 2015: 21-V-101C failed with 4 crack indications. Previous 2012 inspection was clean, 3 years crack propagation from below detection limit.
- 21-V-101C and 21-V-201B are inactive
- Anomaly growth monitoring 6 active vessels (21-V-101 A/B/D and 21-V-201 A/C/D) under 4-monthly PAUT from Aug 2016:
 - New defect found in Apr 2019: 21-V-201A (1 new defect), 21-V-201C (1 new defect).
 - New defect found in Dec 2020: 21-V-201C (1 new defect).
 - July 2022 to Jan 2025: no anomaly growth (correlates with flowthrough operation).
- Current defect sizes remain acceptable per the 2017 critical/tolerable flaw size curves (derived from 21-V-201A FFS). No growth over the ~2.5-year flowthrough period.

6.2 Decision Matrix Criticality & RBI Risk Summary

Table 6-2 summarizes the general 2027 condition of the static equipment population by mapping RUL-based criticality against RBI risk category. That table covers components directly impacted, indirectly impacted and not impacted by post tie-in condition. Overall, no items are classified as Criticality 1, while one static component is classified as Criticality 2 and 16 items are classified as Criticality 3, indicating the need for prioritized verification and closer monitoring. Most components fall under Criticality 4, meaning the majority of the population is expected to have sufficient remaining useful life to support the 2040 life extension target, provided that the existing PCMS-RBI inspection and integrity management program is continued. These matrixes also define the priority level of each static component, as shown in Table 3-1, and provide the basis for determining the number of components classified as high priority and low priority according to the decision matrix.

Table 6-2 Decision matrix in Criticality & RBI Risk

Criticality	Static Equipment 2027			
	Low	Medium I	Medium II	High
Criticality 1	0	0	0	0
Criticality 2	0	1	0	0
Criticality 3	0	5	7	4
Criticality 4	34	37	71	16

6.3 Static Equipment Component Directly Impacted by Post Tie-in Summary

Summary for RBI Risk (without inspection term) static equipment directly impacted by post tie-in for component level are shown in Figure 6-1.

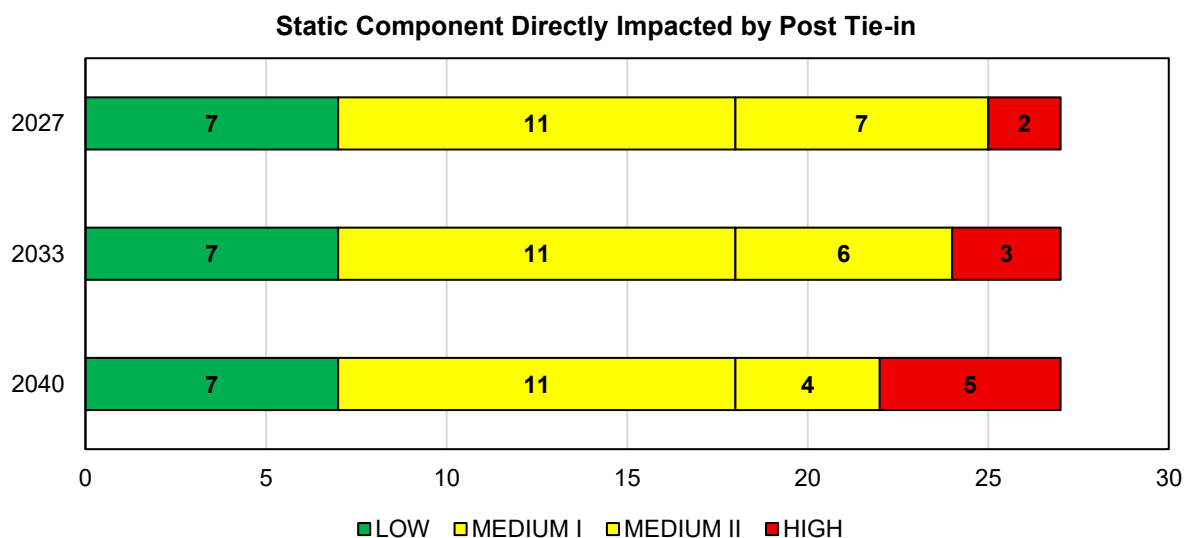


Figure 6-1 RBI Risk static component direct impacted by post tie-in summary

Based on Figure 6-1, the high-risk category without inspection RBI risk term for static equipment components shows an increasing trend. In 2027, two components are classified as high risk and increased by 3 into five static equipment components by 2040. Summary for RUL static equipment directly impacted by post tie-in for component level and equipment level are shown in Table 6-3 and

Table 6-4, respectively.

Table 6-3 RUL static component directly impacted by post tie-in summary

Criticality	RUL definition	Number of Component	Percentage of Component (%)
Criticality 1	RUL 2026 $\leq$ 0 yr	0	0
Criticality 2	0 < RUL 2026 $\leq$ 6 yr (to 2032)	0	0
Criticality 3	6 < RUL 2026 $\leq$ 14 yr (to 2040)	2	7.41
Criticality 4	RUL 2026 > 14 yr	25	92.59
Total		27	100

Table 6-4 RUL static equipment directly impacted by post tie-in summary

Criticality	RUL definition	Number of Equipment	Percentage of Equipment (%)
Criticality 1	RUL 2026 $\leq$ 0 yr	0	0
Criticality 2	0 < RUL 2026 $\leq$ 6 yr (to 2032)	0	0
Criticality 3	6 < RUL 2026 $\leq$ 14 yr (to 2040)	2	11.11
Criticality 4	RUL 2026 > 14 yr	16	88.89
Total		18	100

Table 6-5 Decision matrix for static component directly impacted by post tie-in

Criticality	Static Equipment 2027 (Directly Impacted)			
	Low	Medium I	Medium II	High
Criticality 1	0	0	0	0
Criticality 2	0	0	0	0
Criticality 3	0	2	0	0
Criticality 4	7	9	7	2

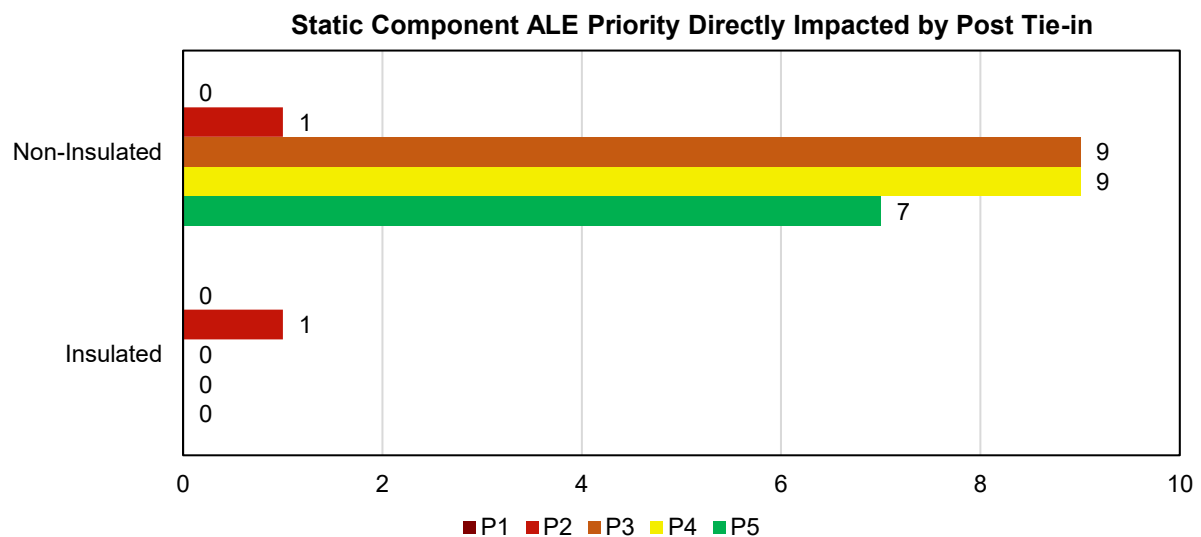


Figure 6-2 Static component directly impacted by post tie-in 2027 ALE Priority

Based on the Table 6-3 and

Table 6-4, at component level Table 6-3 shows that 25 out of 27 components, or 92.59%, have RUL greater than 14 years, while only 2 components, or 7.41%, are classified as Criticality 3 with RUL between 6 and 14 years. At equipment level,

Table 6-4 where 16 out of 18 equipment items, or 88.89%, are classified as Criticality 4, and 2 equipment items, or 11.11%, are classified as Criticality 3. No components or equipment are classified as Criticality 1 or Criticality 2. Therefore, all 27 directly impacted components are projected to reach 2040, 2 of 27 which is classified as Criticality 3 should be verified through review of thickness data, corrosion rate assumptions, and inspection history. On the Table 6-5 and Figure 6-2 shows that most Criticality on 9 components are classified as Priority 3 and 2 components are classified as Priority 2. Therefore, prioritization should not be based only on RUL criticality but also consider the associated RBI risk level. Components with higher ALE priority categories should be prioritized for verification and monitoring to ensure continued fitness for service and alignment with the 2040 life extension objective.

6.4 Static Equipment Indirectly Impacted by Post Tie-in Summary

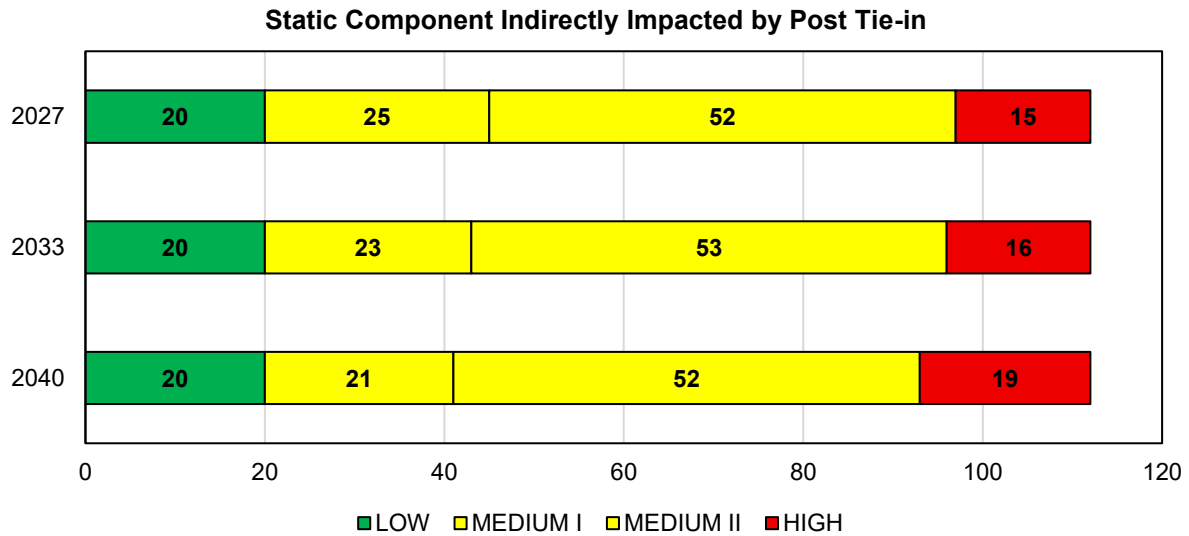


Figure 6-3 RBI Risk static component indirectly impacted by post tie-in summary

Based on Figure 6-3, the high-risk category without inspection RBI risk term for static equipment components indirectly impacted by post tie-in shows an increasing trend. In 2027, 15 (fifteen) components are classified as high risk and increased into 19 (nine-teen) static components in 2040.

Table 6-6 RUL static component indirectly impacted by post tie-in summary

Criticality	RUL definition	Number of Component	Percentage of Component (%)
Criticality 1	$RUL\ 2026 \leq 0\ yr$	0	0
Criticality 2	$0 < RUL\ 2026 \leq 6\ yr\ (to\ 2032)$	1	0.89
Criticality 3	$6 < RUL\ 2026 \leq 14\ yr\ (to\ 2040)$	14	12.5
Criticality 4	$RUL\ 2026 > 14\ yr$	97	86.61
Total		112	100

Table 6-7 RUL static equipment indirectly impacted by post tie-in summary

Criticality	RUL definition	Number of Equipment	Percentage of Equipment (%)
Criticality 1	$RUL\ 2026 \leq 0\ yr$	0	0
Criticality 2	$0 < RUL\ 2026 \leq 6\ yr\ (to\ 2032)$	1	1.23
Criticality 3	$6 < RUL\ 2026 \leq 14\ yr\ (to\ 2040)$	10	12.35
Criticality 4	$RUL\ 2026 > 14\ yr$	70	86.42
Total		81	100

Table 6-8 Decision matrix for static component indirectly impacted by post tie-in

Criticality	Static Equipment 2027 (Indirectly Impacted)			
	Low	Medium I	Medium II	High
Criticality 1	0	0	0	0
Criticality 2	0	1	0	0
Criticality 3	0	3	7	4
Criticality 4	20	21	45	11

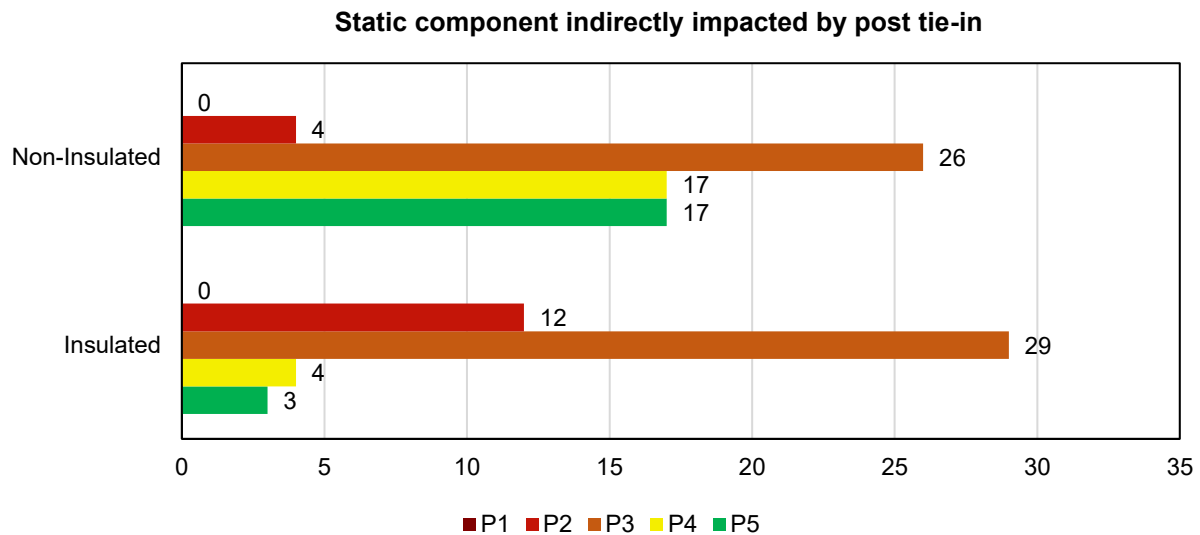


Figure 6-4 Static equipment indirectly impacted by post tie-in ALE priority

Based on Table 6-6 and Table 6-7, the static equipment indirectly impacted by the post tie-in condition is predominantly classified as Criticality 4, indicating that most components and equipment have a RUL exceeding 14 years and are expected to remain fit-for-service through the 2040 life extension target. At the component level, Table 6-6 shows that 97 of 112 components (86.61%) are classified as Criticality 4, while 14 components (12.50%) are classified as Criticality 3 and only one component (35-V-101 SHELL) is classified as Criticality 2. At the equipment level as shown on Table 6-7 indicates that 70 of 81 equipment items (86.42%) are classified as Criticality 4, 10 equipment items (12.35%) are classified as Criticality 3, and one equipment item (1.23%) is classified as Criticality 2. No components or equipment are classified as Criticality 1. Figure 6-4 shows 55 static components with indirectly impacted by post tie-in are fall in ALE Priority 3, reflecting the need for periodic monitoring and verification throughout the life extension period, while 16 static components are categorized as Priority 2 due to their higher risk profile. No components are classified as Priority 1. Therefore, prioritization should not be based solely on RUL criticality but should also consider the associated RBI risk level. Components assigned higher ALE priority categories should be prioritized for inspection, verification, and integrity monitoring to ensure continued fitness for service and support the 2040 life extension objective.

6.5 Static Equipment Component Not Impacted by Post Tie-in Summary

Summary for RBI Risk static equipment not impacted by Post Tie-in for component level are shown in Figure 6-5.

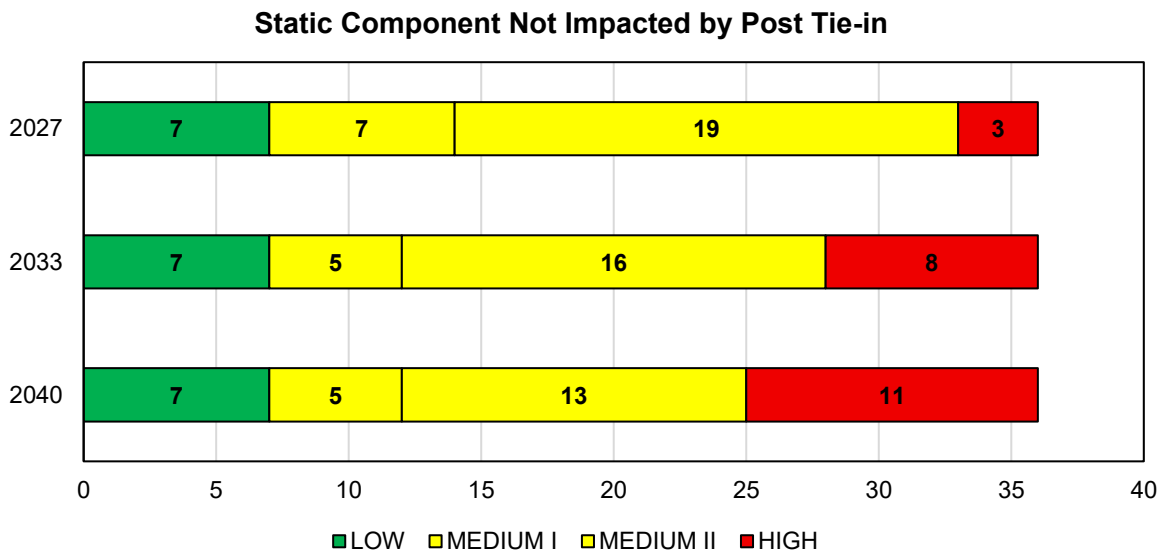


Figure 6-5 RBI Risk static component not impacted by post tie-in summary

Based on Figure 6-5, the high-risk category without the inspection RBI risk term for static equipment components indicates an upward trend. The number of components classified as high risk increases from 3 (three) in 2027 to 11 (eleven) by 2040.

6.6 RUL Static Equipment Not Impacted by Post Tie-in Summary

Summary for RUL static equipment not impacted by Post Tie-in for component level and equipment level are shown in Table 6-9 and Table 6-10, respectively.

Table 6-9 RUL static component not impacted by post tie-in summary

Criticality	RUL definition	Number of Component	Percentage of Component (%)
Criticality 1	RUL 2026 ≤ 0 yr	0	0
Criticality 2	0 < RUL 2026 ≤ 6 yr (to 2032)	0	0
Criticality 3	6 < RUL 2026 ≤ 14 yr (to 2040)	0	0
Criticality 4	RUL 2026 > 14 yr	36	100
Total		36	100

Table 6-10 RUL static equipment not impacted by post tie-in summary

Criticality	RUL definition	Number of Equipment	Percentage of Equipment (%)
Criticality 1	RUL 2026 ≤ 0 yr	0	0
Criticality 2	0 < RUL 2026 ≤ 6 yr (to 2032)	0	0
Criticality 3	6 < RUL 2026 ≤ 14 yr (to 2040)	0	0
Criticality 4	RUL 2026 > 14 yr	33	100
Total		33	100

Table 6-11 Decision matrix for static component not impacted by post tie-in

Criticality	Static Equipment 2027 (Not Impacted)			
	Low	Medium I	Medium II	High
Criticality 1	0	0	0	0
Criticality 2	0	0	0	0
Criticality 3	0	0	0	0
Criticality 4	7	7	19	3

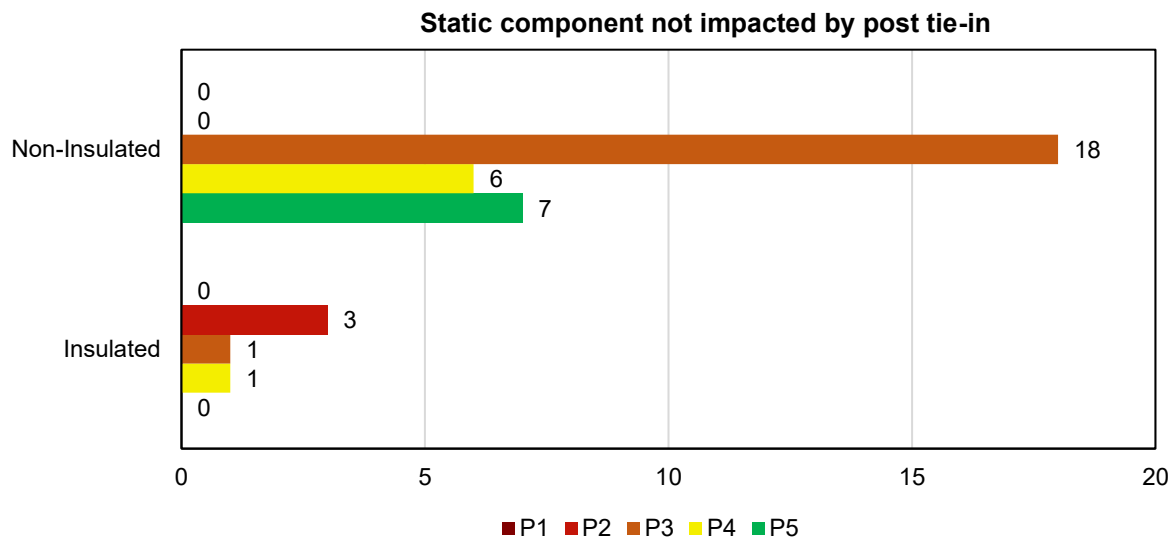


Figure 6-6 Static component not impacted by post tie-in 2027 ALE Priority

Based on Table 6-9 and Table 6-10, the static equipment population not impacted by the post tie-in condition 36 static components or 33 static equipment are classified as Criticality 4. This indicates that most components and equipment have a remaining useful life greater than 14 years and are projected to remain suitable for operation through the 2040 life extension target. In addition to the criticality results, Table 6-11 and Figure 6-6 further supports prioritization by mapping RUL criticality against the RBI risk category to determine the ALE priority. Although all static components are classified as Criticality 4, components with high RBI risk are assigned Priority 2. The decision matrix shows that 19 components are classified as Priority 3 and 3 (three) components are classified as Priority 2. Components within Priority 2 and Priority 3 should be reviewed first to confirm their fitness for continued operation and to determine whether additional inspection, monitoring, or fitness-for-service assessment is required. The complete life extension program list by criticality is provided in Appendix D.

7 CONCLUSIONS AND RECOMMENDATIONS FOR STATIC EQUIPMENT

7.1 Conclusions

The design and operation of 132 equipment items were evaluated. This evaluation covered the following:

1. No asset is classified as Criticality 1 ($RUL \leq 0$ yrs), and no asset has exhausted its remaining wall thickness. No immediate repair or replacement is required based on remaining life.
2. 158 (90.29%) of assessed static components have more than 14 years remaining life, demonstrating strong overall alignment with the 2040 target.
3. 1 static component 35-V-101 shell is classified as Criticality 2 ($6 < RUL \leq 14$ yrs), and 16 static components are classified as Criticality 3 ($6 < RUL \leq 14$ yrs)
4. The summary of 2027 directly impacted by post tie-in condition criticality classification for component level as shown in Table 6-3 is listed below:
 - a. Criticality 1 = 0 component
 - b. Criticality 2 = 0 component
 - c. Criticality 3 = 2 components
 - d. Criticality 4 = 25 components

The summary of 2027 directly impacted by post tie-in condition ALE priority classification for component level as shown in Figure 6-2 is listed below:

- a. Priority 1 = 0 component
 - b. Priority 2 = 2 components
 - c. Priority 3 = 9 components
 - d. Priority 4 = 9 components
 - e. Priority 5 = 7 components
5. The summary of 2027 indirectly by post tie-in condition classification for component level as shown in Table 6-6 is listed below:
 - a. Criticality 1 = 0 component
 - b. Criticality 2 = 1 components
 - c. Criticality 3 = 14 components
 - d. Criticality 4 = 97 components

The summary of 2027 indirectly impacted by post tie-in condition ALE priority classification for component level as shown in Figure 6-4 is listed below:

- a. Priority 1 = 0 component
 - b. Priority 2 = 16 components
 - c. Priority 3 = 55 components
 - d. Priority 4 = 21 components
 - e. Priority 5 = 20 components
6. The summary of 2027 not impacted by post tie-in condition classification for component level as shown in Table 6-9 is listed below:
 - a. Criticality 1 = 0 component
 - b. Criticality 2 = 0 component
 - c. Criticality 3 = 0 components
 - d. Criticality 4 = 36 components

The summary of 2027 not impacted by post tie-in condition classification for component level as shown in Figure 6-6 is listed below:

- a. Priority 1 = 0 component
 - b. Priority 2 = 3 components
 - c. Priority 3 = 19 components
 - d. Priority 4 = 7 components
 - e. Priority 5 = 7 components
7. The 2026 criticality assessment for the post tie-in directly impacted population is dominated by Criticality 4 items (>92%), indicating that most static equipment is projected to remain fit for service through 2040 under the existing PCMS-RBI program.

8. 21 of 175 static components are classified as Priority 2 require near-term inspection. The remaining 154 static components may continue under the existing RBI program toward 2040 target without additional intervention.
9. Inspection priority is driven by both risk and remaining life. Of the 21 static components are classified as Priority 2, 16 static components Criticality 4–High Risk are scheduled for earlier inspection based on risk.
10. The ALE priority assessment indicates that component criticality alone is not the sole basis for determining static equipment integrity priority. Therefore, ALE prioritization shall be established by considering both RUL-based criticality and RBI risk category, rather than relying on RUL criticality alone.

7.2 Recommendations

Based on the current assessment results, the following actions are recommended to support continued safe operation and the ALE extension target:

1. It is important to revisit the RBI assessment every 3 - 5 years as required or when the equipment has operated or is operating beyond the design/operational limitations. Further RBI strategies may be developed in association with a Fitness-For-Service assessment of damaged components using appropriate assessment methods described in API 579-1/ASME FFS-1;
2. Continue the established PCMS-RBI inspection and integrity management program, and incorporate the latest inspection findings, including external inspection, internal inspection, coating inspection, CUI inspection, and IAAAN records, into the RBI review;
3. Use the ALE priority ranking to plan follow-up actions: Priority 2 components should be inspected before the end of 2027, Priority 3 components should be inspected based on half RUL or RBI recommendation, whichever is earlier, and Priority 4 and Priority 5 components should continue under the standing RBI inspection interval and routine monitoring program;
4. Enhance future RBI studies by explicitly incorporating projected future risk profiles for 2033 and 2040. This will provide better visibility of inspection effectiveness and support optimization of inspection intervals and mitigation strategies during the life-extension period
5. Inspect and verify the 21 static components within classified as Priority by end-2027.
6. Verify the predicted simulated post tie-in corrosion rates for Directly Impacted assets against the first available post tie-in thickness inspection data.
7. Update the RBI assessment and ALE priority inspection ranking using the results of Priority 1 and Priority 2 inspections.
8. The operating condition shall be continuously monitored to control the CO₂, H₂S, water cut and pH in the equipment within the specified limit, which could lead to accelerated component deterioration/degradation. Corrective actions shall be taken immediately upon deviation from acceptable limits.
9. Continue the TSA anomaly growth monitoring program.
10. The focus of this study was to manage the risk associated with predictable and inspectable damage mechanisms. However, it does not cover all possible issues, i.e. operational issues/sabotage and damage scenarios, which could lead to a release through small leaks or rupture. The results should be seen as some of the aspects feeding into overall inspection/maintenance plan working alongside other related programs such as:
 - a. Statutory requirement;
 - b. Key process/operation monitoring and control program;
 - c. Monitoring program of the rotating equipment, instrumentation and electrical system on the system;
 - d. Safety Assessment;
 - e. Other special issues including equipment stewardship, root cause failure analysis etc.

APPENDIX A – STATIC EQUIPMENT GENERAL DATA

Static Equipment Directly Impacted by Post tie-in Condition

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pre Tie-In)	Year of Commisioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
19	CGP UNIT 20	Directly Impacted	24-MAJ-105	24-MAJ-105	POLISHING FILTER (CGPX)	GRK 20 - 200/203	2012	1250	175	1111	122	SA 516 Gr.70	20000	N	1	CYL	838.20	40.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
23	CGP UNIT 21	Directly Impacted	21-E-201	21-E-201 SHELL	HEAT EXCHANGER SHELL	GRK 21-7	2000	1250	650	1118	90	SA 516 Gr.70	18800	Y	1	CYL	863.60	40.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + Cladding + CO2 Corrosion + Low Temp H2S Corr.
29	CGP UNIT 21	Directly Impacted	21-S-201	21-S-201	PRODUCT GAS FILTER	GRK 21 - 7	1999	1250	125	1134	86	SA 516 Gr.70	20000	N	1	CYL	749.10	31.75	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + Fatigue - Mechanical
30	CGP UNIT 21	Directly Impacted	21-V-101A	21-V-101-A BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	1999	1250	650	1143	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical
31	CGP UNIT 21	Directly Impacted	21-V-101A	21-V-101-A TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1143	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
32	CGP UNIT 21	Directly Impacted	21-V-101B	21-V-101-B BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	1999	1250	650	1143	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
33	CGP UNIT 21	Directly Impacted	21-V-101B	21-V-101-B TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1143	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Chloride SCC Ext
34	CGP UNIT 21	Directly Impacted	21-V-101D	21-V-101-D BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	2000	1250	650	1143	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical + Chloride SCC Ext
35	CGP UNIT 21	Directly Impacted	21-V-101D	21-V-101-D TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1143	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Chloride SCC Ext
39	CGP UNIT 21	Directly Impacted	21-V-201A	21-V-201-A BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical
40	CGP UNIT 21	Directly Impacted	21-V-201A	21-V-201-A TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
51	CGP UNIT 25	Directly Impacted	24-MAF-401	24-MAF-401 BOTTOM	AMINE CONTACTOR NO. 4 (CGPX)	GRK 25 - 210/211/212/375	2012	1250	200	1085	36	SA 516 Gr.70	20000	N	1	CYL	2209.80	77.00	3.175	External Corrosion Carbon and Low Alloy Steels + Cladding + Chloride SCC Ext
52	CGP UNIT 25	Directly Impacted	24-MAF-401	24-MAF-401 TOP	AMINE CONTACTOR NO. 4 (CGPX)	GRK 25 - 210/211/212/375	2012	1250	200	1085	36	SA 516 Gr.70	20000	N	1	CYL	2209.80	77.00	3.175	External Corrosion Carbon and Low Alloy Steels + Cladding + CO2 Corrosion
53	CGP UNIT 25	Directly Impacted	24-MBD-403	24-MBD-403 BOTTOM	TREATED GAS SEPARATOR NO. 4 (CGPX)	GRK 25 - 381/382	2012	1250	150	1093.4	120	SA 516 Gr.70	20000	N	1	CYL	2210.00	76.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
54	CGP UNIT 25	Directly Impacted	24-MBD-403	24-MBD-403 TOP	TREATED GAS SEPARATOR NO. 4 (CGPX)	GRK 25 - 381/382	2012	1250	150	1093.4	120	SA 516 Gr.70	20000	N	1	CYL	2210.00	76.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
59	CGP UNIT 25	Directly Impacted	25-C-201	25-C-201 BOTTOM	AMINE CONTACTOR BOTTOM	GRK 25 - 306/310	1997	1250	200	1089	170	SA 516 Gr.70	20000	N	1	CYL	2199.89	88.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
60	CGP UNIT 25	Directly Impacted	25-C-201	25-C-201 TOP	AMINE CONTACTOR TOP	GRK 25 - 383/385	1997	1250	200	1084	170	SA 516 Gr.70	20000	N	1	CYL	2199.89	44.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Amine Corrosion + CO2 Corrosion + Low Temp H2S Corr.
63	CGP UNIT 25	Directly Impacted	25-C-301	25-C-301 BOTTOM	AMINE CONTACTOR BOTTOM	GRK 25 - 306/310	1997	1250	200	1128	170	SA 516 Gr.70	20000	N	1	CYL	2199.89	88.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
64	CGP UNIT 25	Directly Impacted	25-C-301	25-C-301 TOP	AMINE CONTACTOR TOP	GRK 25 - 383/385	1997	1250	200	1128	170	SA 516 Gr.70	20000	N	1	CYL	2199.89	88.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Amine Corrosion + CO2 Corrosion
98	CGP UNIT 25	Directly Impacted	25-V-203	25-V-203	TREATED GAS SEPARATOR	GRK 25 - 383/385	1997	1250	150	1093	120	SA 516 Gr.70	20000	N	1	CYL	2209.80	88.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
108	CGP UNIT 25	Directly Impacted	25-V-303	25-V-303	TREATED GAS SEPARATOR	GRK 25 - 384/386	1997	1250	150	1093	120	SA 516 Gr.70	17500	N	1	CYL	2209.80	88.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
128	CGP UNIT 41	Directly Impacted	27-MAF-401	27-MAF-401 BOTTOM	GLYCOL CONTACTOR (CGPX)	GRK 41 - 305/309/377	2012	1250	175	1078	131	SA 516 Gr.70	20000	N	1	CYL	1351.28	47.75	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
129	CGP UNIT 41	Directly Impacted	27-MAF-401	27-MAF-401 TOP	GLYCOL CONTACTOR (CGPX)	GRK 41 - 305/309/377	2012	1250	175	1076	129	SA 516 Gr.70	20000	N	1	CYL	1351.28	47.75	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
131	CGP UNIT 41	Directly Impacted	41-C-201	41-C-201	GLYCOL CONTACTOR	GRK 41 - 3-2/3-3/3-4	1997	1250	175	1092	124	SA 516 Gr.70	20000	N	1	CYL	1350.00	55.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
132	CGP UNIT 41	Directly Impacted	41-C-301	41-C-301	GLYCOL CONTACTOR	GRK 41 - 3-2/3-3/3-4	1997	1250	175	1092	124	SA 516 Gr.70	20000	N	1	CYL	1350.00	55.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
161	CGP UNIT 42	Directly Impacted	42-F-101A	42-F-101A	SALES GAS COALESCING FILTER	GRK 42 - 522	2002	1250	150	1060	115	SA 516 Gr.70	20000	N	1	CYL	1219.00	44.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
162	CGP UNIT 42	Directly Impacted	42-F-101B	42-F-101B	SALES GAS COALESCING FILTER	GRK 42 - 522	2002	1250	150	1060	115	SA 516 Gr.70	20000	N	1	CYL	1219.00	44.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion

Static Equipment Indirectly Impacted by Post tie-in Condition								Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pro Tie-In)	Year of Commissioning	Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
17	CGP UNIT 20	Indirectly Impacted	20-V-17.03	20-V-17.03	POLISHING FILTER	GRK 20 - 200/203	1997	1250	175	1121	120	SA 516 Gr.70	20000	N	1	CYL	838.20	38.10	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
18	CGP UNIT 20	Indirectly Impacted	20-V-17.06	20-V-17.06	POLISHING FILTER	GRK 20 - 200/203	1997	1250	175	1121	120	SA 516 Gr.70	20000	N	1	CYL	837.80	38.10	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S
20	CGP UNIT 20	Indirectly Impacted	24-MAJ-205	24-MAJ-205	POLISHING FILTER (CGPX)	GRK 20 - 200/203	2012	1250	175	1111	122	SA 516 Gr.70	20000	N	1	CYL	838.20	40.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S
21	CGP UNIT 21	Indirectly Impacted	21-E-101	21-E-101 SHELL	HEAT EXCHANGER SHELL	GRK 21-7	2000	1250	650	1118	90	SA 516 Gr.70	18800	Y	1	CYL	863.60	40.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S
22	CGP UNIT 21	Indirectly Impacted	21-E-102	21-E-102 SHELL	GAS COOLER SHELL	GRK 21-14	2000	1250	650	1128	317	SA 516 Gr.70	18800	Y	1	CYL	539.80	25.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S
24	CGP UNIT 21	Indirectly Impacted	21-E-202	21-E-202 SHELL	GAS COOLER SHELL	GRK 21-14	2000	1250	650	1128	317	SA 516 Gr.70	18800	Y	1	CYL	539.80	25.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + Ext. Corr. Carbon and Low
25	CGP UNIT 21	Indirectly Impacted	21-H-101	21-H-101 COMBUSTION	REGENERATION GAS HEATER COMBUSTION SECTION	GRK 50 - 685/686	1997	1250	650	1120	599	N/A	N/A	N	N/A	CYL	N/A	N/A	N/A	High Temperature H2/H2S Corrosion + HIC SOHIC H2S
26	CGP UNIT 21	Indirectly Impacted	21-H-201	21-H-201 COMBUSTION	REGENERATION GAS HEATER COMBUSTION SECTION	GRK 50 - 685/686	1997	1250	650	1120	599	N/A	N/A	N	N/A	CYL	N/A	N/A	N/A	High Temperature H2/H2S Corrosion + HIC SOHIC H2S
27	CGP UNIT 21	Indirectly Impacted	21-S-101	21-S-101	PRODUCT GAS FILTER	GRK 21 - 7	1999	1250	125	1104	85	SA 516 Gr.70	20000	N	1	CYL	749.10	31.75	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S
28	CGP UNIT 21	Indirectly Impacted	21-S-102	21-S-102	COALESCING FILTER	GRK 21 - 101/102	1999	1250	350	1151.7	95	SA 516 Gr.70	20000	N	1	CYL	749.30	31.75	3.175	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
36	CGP UNIT 21	Indirectly Impacted	21-V-102	21-V-102 BOTTOM	HP SEPARATOR BOTTOM	GRK 21 - 22	2000	1250	350	1050	95	SA 516 Gr.70	20000	N	1	CYL	742.75	34.93	3.175	External Corrosion Carbon and Low Alloy Steels + Chloride SCC Ext
37	CGP UNIT 21	Indirectly Impacted	21-V-102	21-V-102 TOP	HP SEPARATOR TOP	GRK 21 - 22	2000	1250	350	1096	95	SA 516 Gr.70	20000	N	1	CYL	742.75	34.93	3.175	External Corrosion Carbon and Low Alloy Steels
38	CGP UNIT 21	Indirectly Impacted	21-V-103	21-V-103	Instrument Air Receiver	GRK 21 - Instrument Air	1999	150	150	115	113	SA 516 Gr.70	20000	N	1	CYL	1200.15	9.53	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
41	CGP UNIT 21	Indirectly Impacted	21-V-201C	21-V-201-C BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical
42	CGP UNIT 21	Indirectly Impacted	21-V-201C	21-V-201-C TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
43	CGP UNIT 21	Indirectly Impacted	21-V-201D	21-V-201-D BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical + CUI Carbon and Low Alloy
44	CGP UNIT 21	Indirectly Impacted	21-V-201D	21-V-201-D TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + CUI Carbon and Low Alloy
45	CGP UNIT 21	Indirectly Impacted	21-V-202	21-V-202 BOTTOM	HP SEPARATOR BOTTOM	GRK 21 - 22	2000	1250	650	1096	95	SA 516 Gr.70	20000	N	1	CYL	742.75	34.93	3.175	External Corrosion Carbon and Low Alloy Steels + HIC SOHIC H2S
46	CGP UNIT 21	Indirectly Impacted	21-V-202	21-V-202 TOP	HP SEPARATOR TOP	GRK 21 - 22	2000	1250	650	1096	95	SA 516 Gr.70	20000	N	1	CYL	742.95	34.93	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
48	CGP UNIT 25	Indirectly Impacted	24-HAL-201	24-HAL-201	TREATED GAS #2 COOLER	GRK 25 - 383/385	2012	1250	200	1084	136	SA 240 Gr 316L	16679	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + Ext. Corr. Carbon and Low
49	CGP UNIT 25	Indirectly Impacted	24-HAL-301	24-HAL-301	TREATED GAS #3 COOLER	GRK 25 - 383/385	2012	1250	200	1084	136	SA 240 Gr 316L	16679	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + HIC SOHIC H2S
50	CGP UNIT 25	Indirectly Impacted	24-HAL-401	24-HAL-401	TREATED GAS #4 COOLER	GRK 25 - 381/382	2012	1250	200	1084	136	SA 240 Gr 316L	16679	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + CO2 Corrosion
57	CGP UNIT 25	Indirectly Impacted	25-C-102	25-C-102 BOTTOM	AMINE REGENERATOR BOTTOM	GRK 25-347A	1997	75	300	16	259	SA 516 Gr.70 + TP304 Clad	20000	Y	1	CYL	4200.00	14.20	3.175	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr.
58	CGP UNIT 25	Indirectly Impacted	25-C-102	25-C-102 TOP	AMINE REGENERATOR TOP	GRK 25-347A	1997	75	300	16	259	SA 516 Gr.70 + TP304 Clad	20000	Y	1	CYL	2860.00	16.20	3.175	CUI Carbon and Low Alloy Steels + CO2 Corrosion
61	CGP UNIT 25	Indirectly Impacted	25-C-202	25-C-202 BOTTOM	AMINE REGENERATOR BOTTOM	GRK 25 - 326/331	1997	75	300	16	259	SA 516 Gr.70 + TP304 Clad	20000	N	1	CYL	4200.00	14.20	3.175	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr. + Chloride SCC Ext
62	CGP UNIT 25	Indirectly Impacted	25-C-202	25-C-202 TOP	AMINE REGENERATOR TOP	GRK 25 - 326/331	1997	75	300	14	120	SA 516 Gr.70	20000	N	1	CYL	2860.00	16.20	3.175	External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion
65	CGP UNIT 25	Indirectly Impacted	25-E-101	25-E-101	REGENERATOR REFLUX CONDENSER	GRK 25 - 332A/B	1997	75	250	14	209	SA 240 304	17500	N	0.85	BOX	3762 (L) x 230 (W) x 152 (H)	10.00	0	Chloride Stress Corrosion Cracking - External + Unspecified Int. Corr.
66	CGP UNIT 25	Indirectly Impacted	25-E-102 A	25-E-102A SHELL	REGENERATOR REBOILER SHELL	GRK 25-345	1997	75	300	16	257	SA 516 Gr.70	20000	Y	0.85	CYL	2547.00	13.20	6.4	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion
67	CGP UNIT 25	Indirectly Impacted	25-E-102 B	25-E-102B SHELL	REGENERATOR REBOILER SHELL	GRK 25-345	1997	75	300	16	257	SA 516 Gr.70	20000	Y	1	CYL	2547.00	16.00	6.4	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion
68	CGP UNIT 25	Indirectly Impacted	25-E-102C	25-E-102C SHELL	REGENERATOR REBOILER SHELL	GRK 25-345	1997	75	300	16	257	SA 516 Gr.70	20000	Y	1	CYL	2547.00	16.50	3.175	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temp H2S Corr.

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pro Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
69	CGP UNIT 25	Indirectly Impacted	25-E-102D	25-E-102D SHELL	REGENERATOR REBOILER SHELL	GRK 25-345	1997	75	300	16	257	SA 516 Gr.70	20000	Y	1	CYL	2547.00	16.45	6.4	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + HIC SOHIC H2S
70	CGP UNIT 25	Indirectly Impacted	25-E-103	25-E-103	LEAN AMINE COOLER	GRK 25 - 357/359	1997	180	250	75	223	SA 240 304	17500	N	0.85	BOX	3000 (L) x 459(W) x 152 (H)	10.00	3	External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Chloride SCC Ext
71	CGP UNIT 25	Indirectly Impacted	25-E-201	25-E-201	REGENERATOR REFLUX CONDENSER	GRK 25 - 332A/B	1997	75	250	14	209	SA 240 304	20000	N	1	BOX	3762 (L) x 152 (W) x 230 (H)	10.00	3.18	Chloride Stress Corrosion Cracking - External
72	CGP UNIT 25	Indirectly Impacted	25-E-202A	25-E-202A SHELL	REGENERATOR REBOILER SHELL	GRK 25-346B	1997	75	300	16	257	SA 516 Gr.70	20000	Y	1	CYL	2517.00	15.00	3.175	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temp H2S Corr.
73	CGP UNIT 25	Indirectly Impacted	25-E-202C	25-E-202C SHELL	REGENERATOR REBOILER SHELL	GRK 25-346B	1997	75	300	16	257	SA 516 Gr.70	17500	Y	0.85	CYL	2513.00	17.00	3.175	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion
74	CGP UNIT 25	Indirectly Impacted	25-E-202B	25-E-202B SHELL	REGENERATOR REBOILER SHELL	GRK 25-346B	1997	75	300	16	257	SA 516 Gr.70	20000	Y	0.85	CYL	2517.00	15.00	6.35	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + HIC SOHIC H2S
75	CGP UNIT 25	Indirectly Impacted	25-E-202D	25-E-202D SHELL	REGENERATOR REBOILER SHELL	GRK 25-346B	1997	75	300	16	257	SA 516 Gr.70	17500	Y	0.85	CYL	2513.00	17.00	6.35	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Unspecified Int. Corr.
76	CGP UNIT 25	Indirectly Impacted	25-E-203	25-E-203	LEAN AMINE COOLER	GRK 25 - 357/359	1997	180	250	75	211	SA 516 Gr.70	20000	N	1	BOX	3000 (L) x 459 (W) x 152 (H)	10.25	3	External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + CUI Carbon and Low Alloy
77	CGP UNIT 25	Indirectly Impacted	25-S-101	25-S-101	AMINE CARTRIDGE FILTER	GRK 25 - 362A/B	1997	180	150	180	120	SA 106 Gr.B	17100	N	1	CYL	590.60	9.50	3.18	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + CUI Carbon and Low Alloy
78	CGP UNIT 25	Indirectly Impacted	25-S-102	25-S-102	AMINE CHARCOAL FILTER	GRK 25 - 374	1997	180	150	50	120	SA 516 Gr.70	20000	N	0.85	CYL	2921.00	22.00	3.18	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + CUI Carbon and Low Alloy
79	CGP UNIT 25	Indirectly Impacted	25-S-201	25-S-201	AMINE CARTRIDGE FILTER	GRK 25 - 362A/B	1997	180	150	50	120	SA 106 Gr.B	17100	N	1	CYL	587.90	9.50	3.18	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
80	CGP UNIT 25	Indirectly Impacted	25-V-101	25-V-101 BOTTOM	REGENERATOR REFLUX ACCUMULATOR BOTTOM	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	16520	N	1	CYL	2032.00	9.00	1.588	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion
81	CGP UNIT 25	Indirectly Impacted	25-V-101	25-V-101 TOP	REGENERATOR REFLUX ACCUMULATOR TOP	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	16520	N	1	CYL	2032.00	9.00	1.588	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion
82	CGP UNIT 25	Indirectly Impacted	25-V-102	25-V-102 BOTTOM	RICH AMINE FLASH DRUM BOTTOM	GRK 25-320	1998	195	400	100	167	SA 516 Gr.70	20000	Y	1	CYL	3048.00	18.00	0	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion
83	CGP UNIT 25	Indirectly Impacted	25-V-102	25-V-102 TOP	RICH AMINE FLASH DRUM TOP	GRK 25-320	1998	195	400	100	167	SA 516 Gr.70	20000	Y	1	CYL	3048.00	18.00	0	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion
86	CGP UNIT 25	Indirectly Impacted	25-V-105A	25-V-105A TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	0	CUI Carbon and Low Alloy Steels
87	CGP UNIT 25	Indirectly Impacted	25-V-105B	25-V-105B BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
88	CGP UNIT 25	Indirectly Impacted	25-V-105B	25-V-105B TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
89	CGP UNIT 25	Indirectly Impacted	25-V-105C	25-V-105C BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
90	CGP UNIT 25	Indirectly Impacted	25-V-105C	25-V-105C TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels
91	CGP UNIT 25	Indirectly Impacted	25-V-105D	25-V-105D BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
92	CGP UNIT 25	Indirectly Impacted	25-V-105D	25-V-105D TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
93	CGP UNIT 25	Indirectly Impacted	25-V-106	25-V-106	MAKE-UP STEAM KO POT	GRK 25-351A	1997	195	400	150	366	SA 516 Gr.70	20000	Y	1	CYL	508.00	9.53	3.18	Unspecified Internal Corrosion + CUI Carbon and Low Alloy
94	CGP UNIT 25	Indirectly Impacted	25-V-201	25-V-201 BOTTOM	REGENERATOR REFLUX ACCUMULATOR BOTTOM	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	20000	N	1	CYL	2032.00	9.00	1.6	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion
95	CGP UNIT 25	Indirectly Impacted	25-V-201	25-V-201 TOP	REGENERATOR REFLUX ACCUMULATOR TOP	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	20000	N	1	CYL	2032.00	9.00	1.6	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion + HIC SOHIC H2S
96	CGP UNIT 25	Indirectly Impacted	25-V-202	25-V-202 BOTTOM	RICH AMINE FLASH DRUM BOTTOM	GRK 25-320	1997	150	200	100	167	SA 516 Gr.70	20000	Y	1	CYL	3048.00	18.00	3.2	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + HIC SOHIC H2S
97	CGP UNIT 25	Indirectly Impacted	25-V-202	25-V-202 TOP	RICH AMINE FLASH DRUM TOP	GRK 25-320	1997	150	200	100	167	SA 516 Gr.70	20000	Y	1	CYL	3048.00	18.00	3.2	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
99	CGP UNIT 25	Indirectly Impacted	25-V-205A	25-V-205A BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
100	CGP UNIT 25	Indirectly Impacted	25-V-205A	25-V-205A TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
101	CGP UNIT 25	Indirectly Impacted	25-V-205B	25-V-205B BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pro Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
102	CGP UNIT 25	Indirectly Impacted	25-V-205B	25-V-205B TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
103	CGP UNIT 25	Indirectly Impacted	25-V-205C	25-V-205C BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
104	CGP UNIT 25	Indirectly Impacted	25-V-205C	25-V-205C TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr.
105	CGP UNIT 25	Indirectly Impacted	25-V-205D	25-V-205D BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
106	CGP UNIT 25	Indirectly Impacted	25-V-205D	25-V-205D TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr.
107	CGP UNIT 25	Indirectly Impacted	25-V-206	25-V-206	MAKE-UP STEAM KO POT	GRK 25-351B	1997	195	400	150	366	SA 516 Gr.70	20000	Y	1	CYL	508.00	9.53	3.18	Unspecified Internal Corrosion + CUI Carbon and Low Alloy
109	CGP UNIT 25	Indirectly Impacted	46-MAJ-001	46-MAJ-001	POST AMINE CARTRIDGE FILTER	GRK 25 - 362A/B	2011	180	150	50	120	SA 106 Gr.B	20000	N	1	CYL	437.00	10.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
119	CGP UNIT 35	Indirectly Impacted	35-C-101	35-C-101 BOTTOM	CONDENSATE STABILIZER BOTTOM	GRK 35-455	1997	200	365	87	316	SA 516 Gr.70	20000	Y	1	CYL	823.00	8.00	1.6	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
120	CGP UNIT 35	Indirectly Impacted	35-C-101	35-C-101 TOP	CONDENSATE STABILIZER TOP	GRK 35-4512	1997	200	365	87	316	SA 516 Gr.70	20000	Y	1	CYL	823.00	8.00	1.6	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
121	CGP UNIT 35	Indirectly Impacted	35-E-102	35-E-102 CHANNEL	CONDENSATE STABILIZER REBOILER CHANNEL	GRK 35-351A/351B	2015	200	365	150	290	SA 516 Gr.70	20000	Y	1	CYL	550.00	10.00	1.5875	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
122	CGP UNIT 35	Indirectly Impacted	35-E-102	35-E-102 SHELL	CONDENSATE STABILIZER REBOILER SHELL	GRK 35-351A/351B	2015	200	365	87	194	SA 516 Gr.70	20000	Y	1	CYL	927.00	12.00	1.5875	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
123	CGP UNIT 35	Indirectly Impacted	35-V-101	35-V-101 BOOT	STABILIZER FEED DRUM BOOT	GRK 35-455	1997	320	95	165	42	SA 516 Gr.70	15000	Y	1	CYL	437.94	9.54	3.175	CUI Carbon and Low Alloy Steels + CO2 Corrosion
124	CGP UNIT 35	Indirectly Impacted	35-V-101	35-V-101 HEAD A	STABILIZER FEED DRUM TOP	GRK 35-455	1997	320	95	165	42	SA 516 Gr.70	20000	Y	1	CYL	1524.00	18.00	3.175	CUI Carbon and Low Alloy Steels + CO2 Corrosion
125	CGP UNIT 35	Indirectly Impacted	35-V-101	35-V-101 HEAD B	STABILIZER FEED DRUM BOTTOM	GRK 35-455	1997	320	95	165	42	SA 516 Gr.70	20000	Y	1	CYL	1524.00	18.00	3.175	CUI Carbon and Low Alloy Steels + CO2 Corrosion
126	CGP UNIT 35	Indirectly Impacted	35-V-101	35-V-101 SHELL	STABILIZER FEED DRUM BOTTOM	GRK 35-455	1997	320	95	165	42	SA 516 Gr.70	20000	Y	1	CYL	1524.00	18.00	3.175	CUI Carbon and Low Alloy Steels + CO2 Corrosion
127	CGP UNIT 41	Indirectly Impacted	27-HBG-401	27-HBG-401 SHELL	GAS/LEAN GLYCOL EXCHANGER (CGPX)	GRK 41 - 12-2/12-3/12-4	2012	1152	450	1152	234	SA 516 Gr.70	20000	N	1	CYL	515.00	20.00	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
135	CGP UNIT 41	Indirectly Impacted	41-S-102	41-S-102	GLYCOL CARBON FILTER	GRK 41 - 6-1	1997	150	200	50	171	SA 106 Gr.B	20000	N	0.85	CYL	590.00	9.80	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Chloride SCC Ext
136	CGP UNIT 41	Indirectly Impacted	41-S-201A	41-S-201 A	GLYCOL CATRIDGE FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	75	171	SA 53B	14870	N	0.85	CYL	342.90	6.35	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
137	CGP UNIT 41	Indirectly Impacted	41-S-201B	41-S-201 B	GLYCOL CATRIDGE FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	75	171	SA 53B	14600	N	0.85	CYL	342.90	6.35	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
138	CGP UNIT 41	Indirectly Impacted	41-S-202	41-S-202 BOTTOM	GLYCOL CARBON FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	50	171	SA 516 Gr.70	14892	N	0.85	CYL	590.00	9.80	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
139	CGP UNIT 41	Indirectly Impacted	41-S-202	41-S-202 SHELL	GLYCOL CARBON FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	50	171	SA 106 Gr.B	14892	N	0.85	CYL	590.00	9.80	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
140	CGP UNIT 41	Indirectly Impacted	41-S-202	41-S-202 TOP	GLYCOL CARBON FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	50	171	SA 516 Gr.70	11942	N	0.85	CYL	590.00	9.80	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
141	CGP UNIT 41	Indirectly Impacted	41-S-301A	41-S-301 A	GLYCOL CATRIDGE FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	75	171	SA 53B	17500	N	0.85	CYL	342.90	6.35	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
142	CGP UNIT 41	Indirectly Impacted	41-S-301B	41-S-301 B	GLYCOL CATRIDGE FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	75	171	SA 53B	14870	N	0.85	CYL	342.90	6.35	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
143	CGP UNIT 41	Indirectly Impacted	41-S-302	41-S-302	GLYCOL CARBON FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	50	171	SA 106 Gr.B	15000	N	1	CYL	570.40	9.80	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
147	CGP UNIT 41	Indirectly Impacted	41-V-203	41-V-203	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-2/4-3/4-4	1997	150	200	75	175	SA 516 Gr 60	20000	N	0.85	CYL	1098.42	10.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
148	CGP UNIT 41	Indirectly Impacted	41-V-204	41-V-204	GLYCOL ACCUMULATOR	GRK 41 - 8-2/8-3/8-4	1997	7	450	2	358	SA 516 Gr 60	20000	Y	0.85	CYL	1516.38	7.62	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
149	CGP UNIT 41	Indirectly Impacted	41-V-303	41-V-303	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-2/4-3/4-4	1997	150	200	75	175	SA 516 Gr 60	20000	N	0.85	CYL	1097.28	10.00	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
150	CGP UNIT 41	Indirectly Impacted	41-V-304	41-V-304	GLYCOL ACCUMULATOR	GRK 41 - 8-2/8-3/8-4	1997	7	450	2	358	SA 516 Gr 60	20000	Y	0.85	CYL	1516.38	7.62	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
151	CGP UNIT 41	Indirectly Impacted	47-BAF-402	47-BAF-402 BOTTOM	GLYCOL STILL COLUMN (CGPX)	GRK 41- 9-4	2012	7	450	1.5	214	SA 240 Gr 316L	15250	Y	0.85	CYL	542.80	8.00	0	Chloride Stress Corrosion Cracking - External

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pro Tie-In)	Year of Commisioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
152	CGP UNIT 41	Indirectly Impacted	47-BAF-402	47-BAF-402 TOP	GLYCOL STILL COLUMN (CGPX)	GRK 41 - 3-2/3-3/3-4	2012	7	450	1.5	214	SA 240 Gr 316L	15250	N	0.85	CYL	542.80	8.00	0	Chloride Stress Corrosion Cracking - External + CO2 Corrosion
153	CGP UNIT 41	Indirectly Impacted	47-BBH-404	47-BBH-404	GLYCOL ACCUMULATOR (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	7	450	2	392	SA 516 Gr.70	15000	N	0.85	CYL	1534.16	12.70	3.2	Unspecified Internal Corrosion + CO2 Corrosion
154	CGP UNIT 41	Indirectly Impacted	47-MAJ-401-A	47-MAJ-401-A	GLYCOL CARTRIDGE FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	75	139	SA 106 Gr.B	15000	N	1	CYL	3530.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
155	CGP UNIT 41	Indirectly Impacted	47-MAJ-401-B	47-MAJ-401-B	GLYCOL CARTRIDGE FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	75	139	SA 106 Gr.B	15000	N	1	CYL	3530.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
156	CGP UNIT 41	Indirectly Impacted	47-MAJ-402	47-MAJ-402	GLYCOL CARBON FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	68	139	SA 516 Gr.70	15000	N	1	CYL	692.20	9.50	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
157	CGP UNIT 41	Indirectly Impacted	47-MAJ-403	47-MAJ-403	GLYCOL FINE FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	68	139	SA 106 Gr.B	17100	N	1	CYL	202.76	8.17	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temp H2S Corr.
158	CGP UNIT 41	Indirectly Impacted	47-MBD-403	47-MBD-403 HEAD 1	RICH GLYCOL FLASH SEPARATOR (CGPX)	GRK 41 - 4-2/4-3/4-4	2012	150	200	75	139	SA 516 Gr.70	15000	N	0.85	CYL	1244.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.
159	CGP UNIT 41	Indirectly Impacted	47-MBD-403	47-MBD-403 HEAD 2	RICH GLYCOL FLASH SEPARATOR (CGPX)	GRK 41 - 4-2/4-3/4-4	2012	150	200	75	139	SA 516 Gr.70	15000	N	0.85	CYL	1244.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
160	CGP UNIT 41	Indirectly Impacted	47-MBD-403	47-MBD-403 SHELL	RICH GLYCOL FLASH SEPARATOR (CGPX)	GRK 41 - 4-2/4-3/4-4	2012	150	200	75	139	SA 516 Gr.70	15000	N	0.85	CYL	1244.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
163	CGP UNIT 46	Indirectly Impacted	46-E-101	46-E-101	EXCESS STEAM CONDENSER	GRK 46 - Low Pressure Steam	2018	75	345	5	220	SA 240 304	18300	N	1	BOX	(10916 (L) x 3321 (W) x 869 (H))	12.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
164	CGP UNIT 46	Indirectly Impacted	46-V-105	46-V-105	CONDENSATE DRUM	GRK 46-4515	1997	75	245	3	220	SA 516 Gr.70	20000	Y	0.85	CYL	914.40	8.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
165	CGP UNIT 47	Indirectly Impacted	47-HBG-401	47-HBG-401	LEAN/RICH GLYCOL EXCHANGER	GRK 41- 7-4 and 10-4	2012	150	450	68	348	SA 106 Gr.B	17100	Y	1	CYL	3530.60	12.70	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
166	CGP UNIT 47	Indirectly Impacted	47-S-101	47-S-101 HEAD BOTTOM	FUEL GAS FILTER	GRK 47 - 663	1997	260	150	215	85	SA 516 Gr.70	17500	N	1	CYL	254.46	9.27	1.59	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
167	CGP UNIT 47	Indirectly Impacted	47-S-101	47-S-101 SHELL	FUEL GAS FILTER	GRK 47 - 663	1997	260	150	215	85	SA 106 Gr.B	14946	N	1	CYL	273.00	9.27	1.59	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
168	CGP UNIT 48	Indirectly Impacted	48-V-101	48-V-101 HEAD A	FLARE KO DRUM BOTTOM	GRK 48 - Flare Header	1997	50	350	5	70	SA 516 Gr.70	20000	N	0.85	CYL	5182.00	16.00	3.2	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion

Static Equipment Not Impacted by Post tie-in Condition

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pro Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
1	CGP UNIT 15	Not Impacted	15-F-101A	15-F-101 A	SUBAN P/L FILTER	GRK 15 - 155/156	2002	1400	150	1140	83	SA 516 Gr.70	20000	N	1	CYL	980.80	43.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
2	CGP UNIT 15	Not Impacted	15-F-101B	15-F-101 B	SUBAN P/L FILTER	GRK 15 - 155/156	2002	1400	150	1140	83	SA 516 Gr.70	20000	N	1	CYL	980.80	43.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
3	CGP UNIT 15	Not Impacted	15-F-101C	15-F-101 C	SUBAN P/L FILTER	GRK 15 - 155/156	2006	1400	150	1140	83	SA 516 Gr.70	20000	N	1	CYL	787.40	38.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
4	CGP UNIT 15	Not Impacted	15-M-101	15-M-101A	GLT Inlet Separator	GRK 15 - 107	2000	1350	120	1140	84	SA 516 Gr.70	17500	N	1	CYL	838.20	38.10	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
5	CGP UNIT 15	Not Impacted	15-V-101	15-V-101	DAYUNG PIPELINE PIG RECEIVER	GRK 15 - 155/156	1997	1400	150	1100	130	API 5L X52	20000	N	1	CYL	547.28	30.69	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
6	CGP UNIT 15	Not Impacted	15-V-102	15-V-102	LETANG/TENGAH PIPELINE PIG RECEIVER	GRK 15 - 52B	1997	1400	150	1130	85	API 5L X52	52000	N	1	CYL	889.00	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
7	CGP UNIT 15	Not Impacted	15-V-103	15-V-103	GELAM GAS LINE PIG RECEIVER	GRK 15 - 105	1997	1400	150	1130	85	API 5L X52	52000	N	1	CYL	298.45	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
8	CGP UNIT 15	Not Impacted	15-V-104	15-V-104 BOTTOM	GLT INLET SEPARATOR BOTTOM	GRK 15 - 108	1997	1250	120	176	53	SA 516 Gr.70	20000	N	1	CYL	182.58	15.09	3.175	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
9	CGP UNIT 15	Not Impacted	15-V-104	15-V-104 TOP	GLT INLET SEPARATOR BOTTOM	GRK 15 - 108	1997	1250	120	1130	84	SA 516 Gr.70	20000	N	1	CYL	749.50	31.75	3.175	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
10	CGP UNIT 15	Not Impacted	15-V-105	15-V-105	DAYUNG INLET SEPARATOR	GRK 15 - 99	1997	1250		1100	83	SA 516 Gr.70	20000	N	1	CYL	1371.60	57.15	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
11	CGP UNIT 15	Not Impacted	15-V-106	15-V-106	GELAM LIQUID LINE PIG RECEIVER	GRK 15 - 104	1997	1400	150	1130	85	API 5L X52	52000	N	1	CYL	78.00	5.50	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
12	CGP UNIT 15	Not Impacted	15-V-107	15-V-107	SUMPAL INLET SEPARATOR	GRK 15 - 99	1997	1250	125	1130	83	SA 516 Gr.70	20000	N	1	CYL	1575.00	57.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
13	CGP UNIT 15	Not Impacted	15-V-108	15-V-108	DAYUNG/SUMPAL PIPELINE PIG RECEIVER	GRK 15 - 99	2003	1400	150	1050	83	SA 516 Gr.70	20000	N	1	CYL	766.00	30.00	1.6	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
14	CGP UNIT 15	Not Impacted	15-V-109	15-V-109	CONDENSATE PIG RECEIVER	GRK 15 - 52B	2006	1400	150	1120	150	Carbon Steel	20000	N	1	CYL	209.50	9.00	N/A	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
15	CGP UNIT 15	Not Impacted	3K-KAQ-001	3K-KAQ-001	Tengah/Letang Rawa Pipeline Pig Receiver (CGPX)	GRK 15 - 155/156	2012	1350	150	1140	86	API 5L X52	52000	N	1	CYL	285.14	8.20	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
16	CGP UNIT 15	Not Impacted	TO-KAQ-001	TO-KAQ-001	SUBAN-GRISSIK TRUNKLINE PIG RECEIVER	GRK 15 - 155/156	2013	1440	200	74.46	98	API 5L X70	20000	N	1	CYL	733.45	14.27	0.9906	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
47	CGP UNIT 25	Not Impacted	24-HAL-101	24-HAL-101	TREATED GAS #1 COOLER	GRK 25 - 387	2012	1250	200	1119	130.6	SA 240 Gr 316L	16700	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + Generic Failure Frequency
55	CGP UNIT 25	Not Impacted	25-C-101	25-C-101 BOTTOM	AMINE CONTACTOR BOTTOM	GRK 25 - 317	1997	1250	200	1125	170	SA 516 Gr.70	20000	N	1	CYL	2199.89	88.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.
56	CGP UNIT 25	Not Impacted	25-C-101	25-C-101 TOP	AMINE CONTACTOR TOP	GRK 25 - 387	1997	1250	200	1119	170	SA 516 Gr.70	20000	N	1	CYL	2199.89	44.10	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
84	CGP UNIT 25	Not Impacted	25-V-103	25-V-103	TREATED GAS SEPARATOR	GRK 25 - 384/386	1997	1250	150	1093	120	SA 516 Gr.70	20000	N	1	CYL	2210.00	88.00	0	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion
85	CGP UNIT 25	Not Impacted	25-V-105A	25-V-105A BOTTOM	TREATED GAS SEPARATOR	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr.70	20000	Y	1	CYL	1219.20	12.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion
110	CGP UNIT 30	Not Impacted	30-E-102	30-E-102 SHELL	GAS CHILLER & HC CODENSATE SUBCOOLER SHE	GRK 30-4515	1997	275	120	57	35	SA 516 Gr.70	17500	Y	1	CYL	1085.00	12.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
111	CGP UNIT 30	Not Impacted	30-E-103	30-E-103	PROPANE CONDENSER	GRK 30 - 414	1997	300	155	250	126	SA 516 Gr.70	20000	N	1	BOX	(248 (L) x 243 (W) x 152 (H))	16.25	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
112	CGP UNIT 30	Not Impacted	30-E-105	30-E-105	Propane Condenser	GRK 30 - 410	1997	275	185	250	120	SA 516 Gr.70	20000	N	0.85	BOX	(2543 (L) x 152 (W) x 243 (H))	16.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
113	CGP UNIT 30	Not Impacted	30-V-101	30-V-101	LOW TEMPERATURE SEPARATOR TOP	GRK 30-403	1997	1250	120	1096	52	SA 516 Gr.70	2000	Y	1	CYL	1981.20	78.00	0	CUI Carbon and Low Alloy Steels + CO2 Corrosion
114	CGP UNIT 30	Not Impacted	30-V-102	30-V-102	DEOPROPANIZER OVERHEADS DRUM	GRK 30 - 414	1997	300	150	175	120	SA 516 Gr.70	20000	N	1	CYL	590.94	9.53	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion
115	CGP UNIT 30	Not Impacted	30-V-103	30-V-103	PROPANE STORAGE DRUM	GRK 30 - 414	1997	275	150	240	122	SA 516 Gr.70	20000	N	1	CYL	1676.40	18.00	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
116	CGP UNIT 30	Not Impacted	30-V-105	30-V-105	PROPANE COMPRESSOR SUCTION DRUM	GRK 30-409	1997	275	100	60	38	SA 516 Gr.70	20000	Y	1	CYL	1270.00	12.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
117	CGP UNIT 30	Not Impacted	30-V-106	30-V-106	REFRIGERANT BOILER	GRK 30 - 410	1997	275	150	175	120	SA 516 Gr.70	20000	N	1	CYL	336.55	8.00	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion
118	CGP UNIT 30	Not Impacted	30-V-104	30-V-104	PROPANE ACCUMULATOR	GRK 30 - 407/411	1997	275	150	240	122	SA 516 Gr.70	20000	N	1	CYL	1676.40	18.00	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pro Tie-In)	Year of Commisioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism
130	CGP UNIT 41	Not Impacted	41-C-101	41-C-101	GLYCOL CONTACTOR	GRK 41 - 316	1997	1250	175	1092	124	SA 516 Gr.70	20000	N	1	CYL	1350.00	55.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
133	CGP UNIT 41	Not Impacted	41-S-101A	41-S-101 A	GLYCOL CATRIDGE FILTER	GRK 41 - 6-1	1997	150	200	75	171	SA 53B	15000	N	0.85	CYL	342.90	6.35	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
134	CGP UNIT 41	Not Impacted	41-S-101B	41-S-101 B	GLYCOL CATRIDGE FILTER	GRK 41 - 6-1	1997	150	200	75	171	SA 53B	14600	N	0.85	CYL	342.90	6.35	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Chloride SCC Ext
144	CGP UNIT 41	Not Impacted	41-V-103	41-V-103 HEAD	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-1	1997	150	200	75	175	SA 516 Gr 60	15000	N	0.85	CYL	1098.42	11.20	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
145	CGP UNIT 41	Not Impacted	41-V-103	41-V-103 SHELL	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-1	1997	150	200	75	175	SA 516 Gr 60	15000	N	0.85	CYL	1098.42	9.90	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion
146	CGP UNIT 41	Not Impacted	41-V-104	41-V-104	GLYCOL ACCUMULATOR	GRK 41 - 8-1	1997	7	450	2	370	SA 516 Gr 60	20000	Y	0.85	CYL	1516.38	7.62	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion

APPENDIX B – LATEST RBI ASSESSMENT RESULTS

Static Equipment Directly Impacted by Post tie-in Condition

NO.	Unit	Category	Equipment ID	Circut ID	Equipment Type	Service / Process (Pn Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision															
								Design Pressure (Psi)	Design Temp. (°F)	Op. Pressure (Psi)	Op. Temp. (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMB Damage Mechanism		Inspection Task	RBI Inspection Date	RBI Risk 2027 (POF/COP) Component	RBI Risk 2027 (POF/COP) Equipment	RBI Risk 2033 (POF/COP) Component	RBI Risk 2033 (POF/COP) Equipment	RBI Risk 2040 (POF/COP) Component	Post Tie-In Component Criticality	RBI Risk 2040 (POF/COP) Equipment	Post Tie-In Equipment Criticality	ALE Priority	ALE Recommendation	ALE Recommendation Inspection Year	Remarks
19	CSP UNIT 20	Directly Impacted	24-MAJ-105	24-MAJ-105	POLISHING FILTER (COPK)	GRK 20 - 200/203	2012	1250	175	1111	122	SA 516 Gr 70	20000	N	1	CYL	839.20	40.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion		VI-02, UT-05	15-Aug-2030	MEDIUM II (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM II (20)	HIGH (20)	Criticality 4	HIGH (20)	Criticality 4	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
23	CSP UNIT 21	Directly Impacted	21-E-201	21-V-201-BWELL	HEAT EXCHANGER SHELL	GRK 21-7	2000	1250	650	1118	90	SA 516 Gr 70	18800	Y	1	CYL	863.60	40.00	3.175	CUJ Carbon and Low Alloy Steels + Unspecified Internal Corrosion + Cladding + CO2 Corrosion + Low Temp H2S Corr.		VI-02	15-Jan-2033	HIGH (20)	HIGH (20)	HIGH (20)	HIGH (20)	HIGH (20)	Criticality 4	HIGH (20)	Criticality 4	P2	Urgent Inspection before end of 2027	2027	Although the RBI inspection is not yet due according to the scheduled inspection date, the ALE prioritisation indicates that this equipment should be inspected earlier
29	CSP UNIT 21	Directly Impacted	21-S-201	21-S-201	PREDUCT GAS FILTER	GRK 21 - 7	1999	1250	125	1134	86	SA 516 Gr 70	20000	N	1	CYL	749.10	31.75	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + Fatigue - Mechanical		VI-02, UT-05	5-Aug-2027	LOW (24)	LOW (24)	LOW (24)	LOW (24)	LOW (24)	Criticality 4	LOW (24)	Criticality 4	P4	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date
30	CSP UNIT 21	Directly Impacted	21-V-101A	21-V-101-A BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 415	1999	1250	650	1143	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical		VI-02, UT-05, UT-11	5-Aug-2031	MEDIUM II (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM II (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
31	CSP UNIT 21	Directly Impacted	21-V-101A	21-V-101-A TOP	ABSORBER VESSEL TOP	GRK 21 - 415	2000	1250	650	1143	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Chloride SCC Init		VI-02, UT-05, UT-11	5-Aug-2031	LOW (24)		LOW (24)	MEDIUM II (20)	LOW (24)	Criticality 4	MEDIUM I (20)		P4	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date
32	CSP UNIT 21	Directly Impacted	21-V-101B	21-V-101-B BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 415	1999	1250	650	1143	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Chloride SCC Init		VI-02, UT-05, UT-11	5-Aug-2027	MEDIUM II (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM II (20)	HIGH (20)	Criticality 4	HIGH (20)	Criticality 4	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
33	CSP UNIT 21	Directly Impacted	21-V-101B	21-V-101-B TOP	ABSORBER VESSEL TOP	GRK 21 - 415	2000	1250	650	1143	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical		VI-02, UT-05, UT-11	5-Aug-2027	LOW (24)		LOW (24)	MEDIUM II (20)	LOW (24)	Criticality 4	MEDIUM I (20)		P4	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date
34	CSP UNIT 21	Directly Impacted	21-V-101D	21-V-101-D BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 415	2000	1250	650	1143	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical		VI-02, UT-05, UT-11	5-Aug-2031	HIGH (20)	HIGH (20)	LOW (24)	HIGH (20)	HIGH (20)	Criticality 4	HIGH (20)	Criticality 4	P2	Urgent Inspection before end of 2027	2027	Although the RBI inspection is not yet due according to the scheduled inspection date, the ALE prioritisation indicates that this equipment should be inspected earlier
35	CSP UNIT 21	Directly Impacted	21-V-101D	21-V-101-D TOP	ABSORBER VESSEL TOP	GRK 21 - 415	2000	1250	650	1143	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Chloride SCC Init		VI-02, UT-05, UT-11	5-Aug-2031	LOW (24)		LOW (24)	LOW (24)	LOW (24)	Criticality 4	MEDIUM I (20)		P4	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date
39	CSP UNIT 21	Directly Impacted	21-V-201A	21-V-201-A BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 415	2000	1250	650	1121	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical		VI-02, UT-05, UT-11	5-Aug-2031	MEDIUM II (20)	MEDIUM I (20)	HIGH (20)	HIGH (20)	HIGH (20)	Criticality 4	HIGH (20)	Criticality 4	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
40	CSP UNIT 21	Directly Impacted	21-V-201A	21-V-201-A TOP	ABSORBER VESSEL TOP	GRK 21 - 415	2000	1250	650	1121	86	SA 516 Gr 70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical		VI-02, UT-05, UT-11	5-Aug-2031	LOW (24)		LOW (24)	HIGH (20)	LOW (24)	Criticality 4	MEDIUM I (20)		P4	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date
51	CSP UNIT 25	Directly Impacted	24-MAF-401	24-MAF-401 BOTTOM	AMINE CONTACTOR NO. 4 (COPK)	GRK 25 - 2102112121375	2012	1250	200	1085	36	SA 516 Gr 70	20000	N	1	CYL	2209.80	77.00	3.175	External Corrosion Carbon and Low Alloy Steels + Cladding + Chloride SCC Init		VI-02	15-Mar-2031	MEDIUM I (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 3	MEDIUM I (20)	Criticality 3	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
52	CSP UNIT 25	Directly Impacted	24-MAF-401	24-MAF-401 TOP	AMINE CONTACTOR NO. 4 (COPK)	GRK 25 - 2102112121375	2012	1250	200	1085	36	SA 516 Gr 70	20000	N	1	CYL	2209.80	77.00	3.175	External Corrosion Carbon and Low Alloy Steels + Cladding + CO2 Corrosion		VI-02	15-Mar-2031	MEDIUM I (24)		MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)		P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
53	CSP UNIT 25	Directly Impacted	24-MBD-403	24-MBD-403 BOTTOM	TREATED GAS SEPARATOR NO. 4 (COPK)	GRK 25 - 381382	2012	1250	150	1093.4	120	SA 516 Gr 70	20000	N	1	CYL	2210.00	76.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Cor.		VI-02, UT-05	15-Mar-2029	MEDIUM I (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
54	CSP UNIT 25	Directly Impacted	24-MBD-403	24-MBD-403 TOP	TREATED GAS SEPARATOR NO. 4 (COPK)	GRK 25 - 381382	2012	1250	150	1093.4	120	SA 516 Gr 70	20000	N	1	CYL	2210.00	76.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion		VI-02, UT-05	15-Mar-2029	MEDIUM I (24)		MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)		P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
59	CSP UNIT 25	Directly Impacted	25-C-201	25-C-201 BOTTOM	AMINE CONTACTOR BOTTOM	GRK 25 - 396310	1997	1250	200	1080	170	SA 516 Gr 70	20000	N	1	CYL	2199.80	88.00	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Cor.		VI-02, UT-05	10-Mar-2027	MEDIUM I (24)	MEDIUM II (20)	MEDIUM I (20)	MEDIUM II (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
60	CSP UNIT 25	Directly Impacted	25-C-201	25-C-201 TOP	AMINE CONTACTOR TOP	GRK 25 - 393385	1997	1250	200	1084	170	SA 516 Gr 70	20000	N	1	CYL	2199.80	44.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Amine Corrosion + CO2 Corrosion + Low Temp H2S Corr.		VI-02, UT-05	10-Mar-2027	MEDIUM II (24)		MEDIUM I (20)	MEDIUM II (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)		P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
63	CSP UNIT 25	Directly Impacted	25-C-301	25-C-301 BOTTOM	AMINE CONTACTOR BOTTOM	GRK 25 - 396310	1997	1250	200	1128	170	SA 516 Gr 70	20000	N	1	CYL	2199.80	88.00	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Cor.		VI-02, UT-05, UT-11	5-Aug-2027	MEDIUM I (24)	MEDIUM II (20)	MEDIUM I (20)	MEDIUM II (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
64	CSP UNIT 25	Directly Impacted	25-C-301	25-C-301 TOP	AMINE CONTACTOR TOP	GRK 25 - 393385	1997	1250	200	1128	170	SA 516 Gr 70	20000	N	1	CYL	2199.80	88.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Amine Corrosion + CO2 Corrosion		VI-02, UT-05, UT-11	5-Aug-2027	MEDIUM II (24)		MEDIUM I (20)	MEDIUM II (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)		P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
98	CSP UNIT 25	Directly Impacted	25-V-203	25-V-203	TREATED GAS SEPARATOR	GRK 25 - 384386	1997	1250	150	1093	120	SA 516 Gr 70	20000	N	1	CYL	2209.80	88.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Cor.		VI-02, UT-05	5-Aug-2027	MEDIUM I (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 3	MEDIUM I (20)	Criticality 3	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
108	CSP UNIT 25	Directly Impacted	25-V-303	25-V-303	TREATED GAS SEPARATOR	GRK 25 - 384386	1997	1250	150	1093	120	SA 516 Gr 70	17500	N	1	CYL	2209.80	88.00	3.175	Wet H2S (H2SO4/H2O/Bleeding) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Cor.		VI-02, UT-05, UT-11	5-Aug-2027	MEDIUM I (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
128	CSP UNIT 41	Directly Impacted	27-MAF-401	27-MAF-401 BOTTOM	GLYCOL CONTACTOR (COPK)	GRK 41 - 305309377	2012	1250	175	1078	131	SA 516 Gr 70	20000	N	1	CYL	1381.28	47.75	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion		VI-02, UT-05	15-Jan-2027	MEDIUM I (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
129	CSP UNIT 41	Directly Impacted	27-MAF-401	27-MAF-401 TOP	GLYCOL CONTACTOR (COPK)	GRK 41 - 305309377	2012	1250	175	1076	129	SA 516 Gr 70	20000	N	1	CYL	1381.28	47.75	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion		VI-02, UT-05	15-Jan-2027	MEDIUM I (24)		MEDIUM I (20)		MEDIUM I (20)	Criticality 4	MEDIUM I (20)		P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
131	CSP UNIT 41	Directly Impacted	41-C-201	41-C-201	GLYCOL CONTACTOR	GRK 41 - 3-29-33-4	1997	1250	175	1082	124	SA 516 Gr 70	20000	N	1	CYL	1380.00	55.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion		VI-02, UT-05	5-Aug-2027	MEDIUM II (24)	MEDIUM II (20)	MEDIUM I (20)	MEDIUM II (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P3	1/2 RLA or RBI recommendation which ever earlier	-	Follow RBI Inspection Plan Date
132	CSP UNIT 41	Directly Impacted	41-C-301	41-C-301	GLYCOL CONTACTOR	GRK 41 - 3-29-33-4	1997	1250	175	1082	124	SA 516 Gr 70	20000	N	1	CYL	1380.00	55.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion		VI-02, UT-05	5-Aug-2027	MEDIUM I (24)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	MEDIUM I (20)	Criticality 4	MEDIUM I (20)	Criticality 4	P4	Follow the RBI plan as stated, the routine RBI program	-	Follow RBI Inspection Plan Date
161	CSP UNIT 42	Directly Impacted	42-F-101A	42-F-101A	SALES GAS COALESCING FILTER	GRK 42 - 522	2002	1250	150	1080	115	SA 516 Gr 70	20000	N	1	CYL	1219.00	44.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Cor.		VI-02, UT-05	5-Aug-2027	LOW (24)	LOW (24)	LOW (24)	LOW (24)	LOW (24)	Criticality 4	LOW (24)	Criticality 4	P5	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date
162	CSP UNIT 42	Directly Impacted	42-F-101B	42-F-101B	SALES GAS COALESCING FILTER	GRK 42 - 522	2002	1250	150	1080	115	SA 516 Gr 70	20000	N	1	CYL	1219.00	44.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion		VI-02, UT-05	5-Aug-2031	LOW (24)	LOW (24)	LOW (24)	LOW (24)	LOW (24)	Criticality 4	LOW (24)	Criticality 4	P5	Monitor only, no action beyond the standing RBI interval.	-	Follow RBI Inspection Plan Date

Static Equipment Indirectly Impacted by Post tie-in Condition

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pre Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		PCMS Damage Mechanism	Final Decision					
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)		RBI Risk 2027 (POF/COF) Component	RBI Risk 2027 (POF/COF) Equipment	RBI Risk 2033 (POF/COF) Component	RBI Risk 2033 (POF/COF) Equipment	RBI Risk 2040 (POF/COF) Component	RBI Risk 2040 (POF/COF) Equipment
17	CGP UNIT 20	Indirectly Impacted	20-V-17.03	20-V-17.03	POLISHING FILTER	GRK 20 - 200/203	1997	1250	175	1121	120	SA 516 Gr.70	20000	N	1	CYL	838.20	38.10	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
18	CGP UNIT 20	Indirectly Impacted	20-V-17.06	20-V-17.06	POLISHING FILTER	GRK 20 - 200/203	1997	1250	175	1121	120	SA 516 Gr.70	20000	N	1	CYL	837.80	38.10	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
20	CGP UNIT 20	Indirectly Impacted	24-MAJ-205	24-MAJ-205	POLISHING FILTER (CGPX)	GRK 20 - 200/203	2012	1250	175	1111	122	SA 516 Gr.70	20000	N	1	CYL	838.20	40.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)
21	CGP UNIT 21	Indirectly Impacted	21-E-101	21-E-101 SHELL	HEAT EXCHANGER SHELL	GRK 21-7	2000	1250	650	1118	90	SA 516 Gr.70	18800	Y	1	CYL	863.60	40.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)
22	CGP UNIT 21	Indirectly Impacted	21-E-102	21-E-102 SHELL	GAS COOLER SHELL	GRK 21-14	2000	1250	650	1128	317	SA 516 Gr.70	18800	Y	1	CYL	539.80	25.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
24	CGP UNIT 21	Indirectly Impacted	21-E-202	21-E-202 SHELL	GAS COOLER SHELL	GRK 21-14	2000	1250	650	1128	317	SA 516 Gr.70	18800	Y	1	CYL	539.80	25.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + Ext. Corr. Carbon and Low	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
25	CGP UNIT 21	Indirectly Impacted	21-H-101	21-H-101 COMBUSTION	REGENERATION GAS HEATER COMBUSTION SECTION	GRK 50 - 685/686	1997	1250	650	1120	599	N/A	N/A	N	N/A	CYL	N/A	N/A	N/A	High Temperature H2/H2S Corrosion + HIC SOHIC H2S	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
26	CGP UNIT 21	Indirectly Impacted	21-H-201	21-H-201 COMBUSTION	REGENERATION GAS HEATER COMBUSTION SECTION	GRK 50 - 685/686	1997	1250	650	1120	599	N/A	N/A	N	N/A	CYL	N/A	N/A	N/A	High Temperature H2/H2S Corrosion + HIC SOHIC H2S	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
27	CGP UNIT 21	Indirectly Impacted	21-S-101	21-S-101	PRODUCT GAS FILTER	GRK 21 - 7	1999	1250	125	1104	85	SA 516 Gr.70	20000	N	1	CYL	749.10	31.75	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + HIC SOHIC H2S	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
28	CGP UNIT 21	Indirectly Impacted	21-S-102	21-S-102	COALESCING FILTER	GRK 21 - 101/102	1999	1250	350	1151.7	95	SA 516 Gr.70	20000	N	1	CYL	749.30	31.75	3.175	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
36	CGP UNIT 21	Indirectly Impacted	21-V-102	21-V-102 BOTTOM	HP SEPARATOR BOTTOM	GRK 21 - 22	2000	1250	350	1050	95	SA 516 Gr.70	20000	N	1	CYL	742.75	34.93	3.175	External Corrosion Carbon and Low Alloy Steels + Chloride SCC Ext	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
37	CGP UNIT 21	Indirectly Impacted	21-V-102	21-V-102 TOP	HP SEPARATOR TOP	GRK 21 - 22	2000	1250	350	1096	95	SA 516 Gr.70	20000	N	1	CYL	742.75	34.93	3.175	External Corrosion Carbon and Low Alloy Steels	MEDIUM I (2C)		MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	
38	CGP UNIT 21	Indirectly Impacted	21-V-103	21-V-103	Instrument Air Receiver	GRK 21 - Instrument Air	1999	150	150	115	113	SA 516 Gr.70	20000	N	1	CYL	1200.15	9.53	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
41	CGP UNIT 21	Indirectly Impacted	21-V-201C	21-V-201-C BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
42	CGP UNIT 21	Indirectly Impacted	21-V-201C	21-V-201-C TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	LOW (3A)		LOW (3A)	LOW (3A)	LOW (3A)	
43	CGP UNIT 21	Indirectly Impacted	21-V-201D	21-V-201-D BOTTOM	ABSORBER VESSEL BOTTOM	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Fatigue - Mechanical + CUI Carbon and Low Alloy	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
44	CGP UNIT 21	Indirectly Impacted	21-V-201D	21-V-201-D TOP	ABSORBER VESSEL TOP	GRK 21 - 4/5	2000	1250	650	1121	86	SA 516 Gr.70	18800	N	1	CYL	2336.80	92.96	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + CUI Carbon and Low Alloy	LOW (3A)		LOW (3A)	LOW (3A)	LOW (3A)	
45	CGP UNIT 21	Indirectly Impacted	21-V-202	21-V-202 BOTTOM	HP SEPARATOR BOTTOM	GRK 21 - 22	2000	1250	650	1096	95	SA 516 Gr.70	20000	N	1	CYL	742.75	34.93	3.175	External Corrosion Carbon and Low Alloy Steels + HIC SOHIC H2S	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
46	CGP UNIT 21	Indirectly Impacted	21-V-202	21-V-202 TOP	HP SEPARATOR TOP	GRK 21 - 22	2000	1250	650	1096	95	SA 516 Gr.70	20000	N	1	CYL	742.95	34.93	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)		MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	
48	CGP UNIT 25	Indirectly Impacted	24-HAL-201	24-HAL-201	TREATED GAS #2 COOLER	GRK 25 - 383/385	2012	1250	200	1084	136	SA 240 Gr 316L	16679	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + Ext. Corr. Carbon and Low	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
49	CGP UNIT 25	Indirectly Impacted	24-HAL-301	24-HAL-301	TREATED GAS #3 COOLER	GRK 25 - 383/385	2012	1250	200	1084	136	SA 240 Gr 316L	16679	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + HIC SOHIC H2S	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
50	CGP UNIT 25	Indirectly Impacted	24-HAL-401	24-HAL-401	TREATED GAS #4 COOLER	GRK 25 - 381/382	2012	1250	200	1084	136	SA 240 Gr 316L	16679	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + CO2 Corrosion	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
57	CGP UNIT 25	Indirectly Impacted	25-C-102	25-C-102 BOTTOM	AMINE REGENERATOR BOTTOM	GRK 25-347A	1997	75	300	16	259	SA 516 Gr.70 + TP304 Clad	20000	Y	1	CYL	4200.00	14.20	3.175	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr.	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
58	CGP UNIT 25	Indirectly Impacted	25-C-102	25-C-102 TOP	AMINE REGENERATOR TOP	GRK 25-347A	1997	75	300	16	259	SA 516 Gr.70 + TP304 Clad	20000	Y	1	CYL	2860.00	16.20	3.175	CUI Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (2D)		MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	
61	CGP UNIT 25	Indirectly Impacted	25-C-202	25-C-202 BOTTOM	AMINE REGENERATOR BOTTOM	GRK 25 - 326/331	1997	75	300	16	259	SA 516 Gr.70 + TP304 Clad	20000	N	1	CYL	4200.00	14.20	3.175	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr. + Chloride SCC Ext	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
62	CGP UNIT 25	Indirectly Impacted	25-C-202	25-C-202 TOP	AMINE REGENERATOR TOP	GRK 25 - 326/331	1997	75	300	14	120	SA 516 Gr.70	2													

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pre Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision						
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism	RBI Risk 2027 (POF/COF) Component	RBI Risk 2027 (POF/COF) Equipment	RBI Risk 2033 (POF/COF) Component	RBI Risk 2033 (POF/COF) Equipment	RBI Risk 2040 (POF/COF) Component	RBI Risk 2040 (POF/COF) Equipment
80	CGP UNIT 25	Indirectly Impacted	25-V-101	25-V-101 BOTTOM	REGENERATOR REFLEX ACCUMULATOR BOTTOM	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	16520	N	1	CYL	2032.00	9.00	1.588	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
81	CGP UNIT 25	Indirectly Impacted	25-V-101	25-V-101 TOP	REGENERATOR REFLEX ACCUMULATOR TOP	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	16520	N	1	CYL	2032.00	9.00	1.588	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion	HIGH (3D)		HIGH (3D)		HIGH (3D)	
82	CGP UNIT 25	Indirectly Impacted	25-V-102	25-V-102 BOTTOM	RICH AMINE FLASH DRUM BOTTOM	GRK 25-320	1998	195	400	100	167	SA 516 Gr 70	20000	Y	1	CYL	3048.00	18.00	0	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
83	CGP UNIT 25	Indirectly Impacted	25-V-102	25-V-102 TOP	RICH AMINE FLASH DRUM TOP	GRK 25-320	1998	195	400	100	167	SA 516 Gr 70	20000	Y	1	CYL	3048.00	18.00	0	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
86	CGP UNIT 25	Indirectly Impacted	25-V-105A	25-V-105A TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	0	CUI Carbon and Low Alloy Steels	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
87	CGP UNIT 25	Indirectly Impacted	25-V-105B	25-V-105B BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
88	CGP UNIT 25	Indirectly Impacted	25-V-105B	25-V-105B TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	HIGH (3D)		HIGH (3D)		HIGH (3D)	
89	CGP UNIT 25	Indirectly Impacted	25-V-105C	25-V-105C BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
90	CGP UNIT 25	Indirectly Impacted	25-V-105C	25-V-105C TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
91	CGP UNIT 25	Indirectly Impacted	25-V-105D	25-V-105D BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
92	CGP UNIT 25	Indirectly Impacted	25-V-105D	25-V-105D TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
93	CGP UNIT 25	Indirectly Impacted	25-V-106	25-V-106	MAKE-UP STEAM KO POT	GRK 25-351A	1997	195	400	150	366	SA 516 Gr 70	20000	Y	1	CYL	508.00	9.53	3.18	Unspecified Internal Corrosion + CUI Carbon and Low Alloy	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
94	CGP UNIT 25	Indirectly Impacted	25-V-201	25-V-201 BOTTOM	REGENERATOR REFLEX ACCUMULATOR BOTTOM	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	20000	N	1	CYL	2032.00	9.00	1.6	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
95	CGP UNIT 25	Indirectly Impacted	25-V-201	25-V-201 TOP	REGENERATOR REFLEX ACCUMULATOR TOP	GRK 25 - 332A/B	1997	75	190	15	140	SA 240 304L	20000	N	1	CYL	2032.00	9.00	1.6	Chloride Stress Corrosion Cracking - External + Unspecified Internal Corrosion + HIC SOHIC H2S	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
96	CGP UNIT 25	Indirectly Impacted	25-V-202	25-V-202 BOTTOM	RICH AMINE FLASH DRUM BOTTOM	GRK 25-320	1997	150	200	100	167	SA 516 Gr 70	20000	Y	1	CYL	3048.00	18.00	3.2	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + HIC SOHIC H2S	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
97	CGP UNIT 25	Indirectly Impacted	25-V-202	25-V-202 TOP	RICH AMINE FLASH DRUM TOP	GRK 25-320	1997	150	200	100	167	SA 516 Gr 70	20000	Y	1	CYL	3048.00	18.00	3.2	CUI Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.	HIGH (3D)		HIGH (3D)		HIGH (3D)	
99	CGP UNIT 25	Indirectly Impacted	25-V-205A	25-V-205A BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
100	CGP UNIT 25	Indirectly Impacted	25-V-205A	25-V-205A TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
101	CGP UNIT 25	Indirectly Impacted	25-V-205B	25-V-205B BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
102	CGP UNIT 25	Indirectly Impacted	25-V-205B	25-V-205B TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.175	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
103	CGP UNIT 25	Indirectly Impacted	25-V-205C	25-V-205C BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
104	CGP UNIT 25	Indirectly Impacted	25-V-205C	25-V-205C TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr.	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
105	CGP UNIT 25	Indirectly Impacted	25-V-205D	25-V-205D BOTTOM	REBOILER CONDENSATE POT BOTTOM	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
106	CGP UNIT 25	Indirectly Impacted	25-V-205D	25-V-205D TOP	REBOILER CONDENSATE POT TOP	GRK 25-373A/373B	1997	195	400	50	297	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	3.2	CUI Carbon and Low Alloy Steels + Unspecified Int. Corr.	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
107	CGP UNIT 25	Indirectly Impacted	25-V-206	25-V-206	MAKE-UP STEAM KO POT	GRK 25-351B	1997	195	400	150	366	SA 516 Gr 70	20000	Y	1	CYL	508.00	9.53	3.18	Unspecified Internal Corrosion + CUI Carbon and Low Alloy	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
109	CGP UNIT 25	Indirectly Impacted	46-MAJ-001	46-MAJ-001	POST AMINE CARTRIDGE FILTER	GRK 25 - 362A/B	2011	180	150	50	120	SA 106 Gr B	20000	N	1	CYL	437.00	10.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.	LOW (2A)	LOW (2A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
119	CGP UNIT 35	Indirectly Impacted	35-C-101	35-C-101 BOTTOM	CONDENSATE STABILIZER BOTTOM	GRK 35-455	1997	200	365	87	316	SA 516 Gr 70	20000	Y	1	CYL	823.00	8.00	1.6	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM I (2C)	MEDIUM II (3C)	MEDIUM I (2C)	MEDIUM II (3C)	MEDIUM I (2C)	MEDIUM II (3C)
120	CGP UNIT 35	Indirectly Impacted	35-C-101	35-C-101 TOP	CONDENSATE STABILIZER TOP	GRK 35-4512	1997	200	365	87	316	SA 516 Gr 70	20000	Y	1	CYL	823.00	8.00	1.6	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM II (3C)		MEDIUM II (3C)		MEDIUM II (3C)	

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pre Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision						
								Design Pressure (Psig)	Design Temp. (°F)	Op. Pressure (Psig)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism	RBI Risk 2027 (POF/COF) Component	RBI Risk 2027 (POF/COF) Equipment	RBI Risk 2033 (POF/COF) Component	RBI Risk 2033 (POF/COF) Equipment	RBI Risk 2049 (POF/COF) Component	RBI Risk 2049 (POF/COF) Equipment
141	CGP UNIT 41	Indirectly Impacted	41-S-301A	41-S-301 A	GLYCOL CATRIDGE FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	75	171	SA 53B	17500	N	0.85	CYL	342.90	6.35	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
142	CGP UNIT 41	Indirectly Impacted	41-S-301B	41-S-301 B	GLYCOL CATRIDGE FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	75	171	SA 53B	14870	N	0.85	CYL	342.90	6.35	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM II (3C)	MEDIUM II (3C)
143	CGP UNIT 41	Indirectly Impacted	41-S-302	41-S-302	GLYCOL CARBON FILTER	GRK 41 - 6-2/6-3/6-4	1997	150	200	50	171	SA 106 Gr B	15000	N	1	CYL	570.40	9.80	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (3B)	MEDIUM I (3B)	MEDIUM I (3B)	MEDIUM I (3B)	MEDIUM I (3B)	MEDIUM I (3B)
147	CGP UNIT 41	Indirectly Impacted	41-V-203	41-V-203	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-2/4-3/4-4	1997	150	200	75	175	SA 516 Gr 60	20000	N	0.85	CYL	1098.42	10.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM II (3C)	MEDIUM II (3C)
148	CGP UNIT 41	Indirectly Impacted	41-V-204	41-V-204	GLYCOL ACCUMULATOR	GRK 41 - 8-2/8-3/8-4	1997	7	450	2	358	SA 516 Gr 60	20000	Y	0.85	CYL	1516.38	7.62	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
149	CGP UNIT 41	Indirectly Impacted	41-V-303	41-V-303	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-2/4-3/4-4	1997	150	200	75	175	SA 516 Gr 60	20000	N	0.85	CYL	1097.28	10.00	3.18	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
150	CGP UNIT 41	Indirectly Impacted	41-V-304	41-V-304	GLYCOL ACCUMULATOR	GRK 41 - 8-2/8-3/8-4	1997	7	450	2	358	SA 516 Gr 60	20000	Y	0.85	CYL	1516.38	7.62	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
151	CGP UNIT 41	Indirectly Impacted	47-BAF-402	47-BAF-402 BOTTOM	GLYCOL STILL COLUMN (CGPX)	GRK 41 - 9-4	2012	7	450	1.5	214	SA 240 Gr 316L	15250	Y	0.85	CYL	542.80	8.00	0	Chloride Stress Corrosion Cracking - External	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
152	CGP UNIT 41	Indirectly Impacted	47-BAF-402	47-BAF-402 TOP	GLYCOL STILL COLUMN (CGPX)	GRK 41 - 3-2/3-3/3-4	2012	7	450	1.5	214	SA 240 Gr 316L	15250	N	0.85	CYL	542.80	8.00	0	Chloride Stress Corrosion Cracking - External + CO2 Corrosion	MEDIUM II (3C)		MEDIUM II (3C)		MEDIUM II (3C)	
153	CGP UNIT 41	Indirectly Impacted	47-BBH-404	47-BBH-404	GLYCOL ACCUMULATOR (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	7	450	2	392	SA 516 Gr 70	15000	N	0.85	CYL	1534.16	12.70	3.2	Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
154	CGP UNIT 41	Indirectly Impacted	47-MAJ-401-A	47-MAJ-401-A	GLYCOL CARTRIDGE FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	75	139	SA 106 Gr B	15000	N	1	CYL	3530.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
155	CGP UNIT 41	Indirectly Impacted	47-MAJ-401-B	47-MAJ-401-B	GLYCOL CARTRIDGE FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	75	139	SA 106 Gr B	15000	N	1	CYL	3530.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
156	CGP UNIT 41	Indirectly Impacted	47-MAJ-402	47-MAJ-402	GLYCOL CARBON FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	68	139	SA 516 Gr 70	15000	N	1	CYL	692.20	9.50	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)
157	CGP UNIT 41	Indirectly Impacted	47-MAJ-403	47-MAJ-403	GLYCOL FINE FILTER (CGPX)	GRK 41 - 6-2/6-3/6-4	2012	150	200	68	139	SA 106 Gr B	17100	N	1	CYL	202.76	8.17	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temp H2S Corr.	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)	MEDIUM I (2B)
158	CGP UNIT 41	Indirectly Impacted	47-MBD-403	47-MBD-403 HEAD 1	RICH GLYCOL FLASH SEPARATOR (CGPX)	GRK 41 - 4-2/4-3/4-4	2012	150	200	75	139	SA 516 Gr 70	15000	N	0.85	CYL	1244.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Unspecified Int. Corr.	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
159	CGP UNIT 41	Indirectly Impacted	47-MBD-403	47-MBD-403 HEAD 2	RICH GLYCOL FLASH SEPARATOR (CGPX)	GRK 41 - 4-2/4-3/4-4	2012	150	200	75	139	SA 516 Gr 70	15000	N	0.85	CYL	1244.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (3C)		MEDIUM II (3C)		MEDIUM II (3C)	
160	CGP UNIT 41	Indirectly Impacted	47-MBD-403	47-MBD-403 SHELL	RICH GLYCOL FLASH SEPARATOR (CGPX)	GRK 41 - 4-2/4-3/4-4	2012	150	200	75	139	SA 516 Gr 70	15000	N	0.85	CYL	1244.60	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)		MEDIUM II (3C)		MEDIUM II (3C)	
163	CGP UNIT 46	Indirectly Impacted	46-E-101	46-E-101	EXCESS STEAM CONDENSER	GRK 46 - Low Pressure Steam	2018	75	345	5	220	SA 240 304	18300	N	1	BOX	(10916 (L) x 3321 (W) x 869 (H))	12.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
164	CGP UNIT 46	Indirectly Impacted	46-V-105	46-V-105	CONDENSATE DRUM	GRK 46-4515	1997	75	245	3	220	SA 516 Gr 70	20000	Y	0.85	CYL	914.40	8.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (3A)	LOW (3A)
165	CGP UNIT 47	Indirectly Impacted	47-HBG-401	47-HBG-401	LEANRICH GLYCOL EXCHANGER	GRK 41 - 7-4 and 10-4	2012	150	450	68	348	SA 106 Gr B	17100	Y	1	CYL	3530.60	12.70	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
166	CGP UNIT 47	Indirectly Impacted	47-S-101	47-S-101 HEAD BOTTOM	FUEL GAS FILTER	GRK 47 - 663	1997	260	150	215	85	SA 516 Gr 70	17500	N	1	CYL	254.46	9.27	1.59	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
167	CGP UNIT 47	Indirectly Impacted	47-S-101	47-S-101 SHELL	FUEL GAS FILTER	GRK 47 - 663	1997	260	150	215	85	SA 106 Gr B	14946	N	1	CYL	273.00	9.27	1.59	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	LOW (2A)		LOW (2A)		LOW (2A)	
168	CGP UNIT 48	Indirectly Impacted	48-V-101	48-V-101 HEAD A	FLARE KO DRUM BOTTOM	GRK 48 - Flare Header	1997	50	350	5	70	SA 516 Gr 70	20000	N	0.85	CYL	5182.00	16.00	3.2	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	

Static Equipment Not Impacted by Post tie-in Condition

NO.	Unit	Category	Equipment ID	Circuit ID	Equipment Type	Service / Process (Pre Tie-In)	Year of Commissioning	Design & Operating Data				Material Data		Insulated (Y/N)	Joint Efficiency	Geometry Data		Thickness Data		Final Decision						
								Design Pressure (Psi/g)	Design Temp. (T)	Op. Pressure (Psi/g)	Op. Temp (°F)	Material	Design Stress (Psi)			Geometry	Internal Diameter (mm)	Thickness Nominal (mm)	Corrosion Allow. (mm)	PCMS Damage Mechanism	RBI Risk 2027 (POF/COF) Component	RBI Risk 2027 (POF/COF) Equipment	RBI Risk 2033 (POF/COF) Component	RBI Risk 2033 (POF/COF) Equipment	RBI Risk 2040 (POF/COF) Component	RBI Risk 2040 (POF/COF) Equipment
1	CGP UNIT 15	Not Impacted	15-F-101A	15-F-101 A	SUBAN PIL FILTER	GRK 15 - 155/156	2002	1400	150	1140	83	SA 516 Gr 70	20000	N	1	CYL	980.80	43.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
2	CGP UNIT 15	Not Impacted	15-F-101B	15-F-101 B	SUBAN PIL FILTER	GRK 15 - 155/156	2002	1400	150	1140	83	SA 516 Gr 70	20000	N	1	CYL	980.80	43.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
3	CGP UNIT 15	Not Impacted	15-F-101C	15-F-101 C	SUBAN PIL FILTER	GRK 15 - 155/156	2006	1400	150	1140	83	SA 516 Gr 70	20000	N	1	CYL	787.40	38.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (3A)	LOW (3A)	LOW (3A)	LOW (3A)
4	CGP UNIT 15	Not Impacted	15-M-101	15-M-101A	GLT Inlet Separator	GRK 15 - 107	2000	1350	120	1140	84	SA 516 Gr 70	17500	N	1	CYL	838.20	38.10	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)
5	CGP UNIT 15	Not Impacted	15-V-101	15-V-101	DAYUNG PIPELINE PIG RECEIVER	GRK 15 - 155/156	1997	1400	150	1100	130	API 5L X52	20000	N	1	CYL	547.28	30.69	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
6	CGP UNIT 15	Not Impacted	15-V-102	15-V-102	LETANG/TENGAH PIPELINE PIG RECEIVER	GRK 15 - 52B	1997	1400	150	1130	85	API 5L X52	52000	N	1	CYL	889.00	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
7	CGP UNIT 15	Not Impacted	15-V-103	15-V-103	GELAM GAS LINE PIG RECEIVER	GRK 15 - 105	1997	1400	150	1130	85	API 5L X52	52000	N	1	CYL	298.45	12.70	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
8	CGP UNIT 15	Not Impacted	15-V-104	15-V-104 BOTTOM	GLT INLET SEPARATOR BOTTOM	GRK 15 - 108	1997	1250	120	1176	53	SA 516 Gr 70	20000	N	1	CYL	182.58	15.09	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
9	CGP UNIT 15	Not Impacted	15-V-104	15-V-104 TOP	GLT INLET SEPARATOR BOTTOM	GRK 15 - 108	1997	1250	120	1130	84	SA 516 Gr 70	20000	N	1	CYL	749.50	31.75	3.175	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)		HIGH (3D)		HIGH (3D)	
10	CGP UNIT 15	Not Impacted	15-V-105	15-V-105	DAYUNG INLET SEPARATOR	GRK 15 - 99	1997	1250	-	1100	83	SA 516 Gr 70	20000	N	1	CYL	1371.60	57.15	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
11	CGP UNIT 15	Not Impacted	15-V-106	15-V-106	GELAM LIQUID LINE PIG RECEIVER	GRK 15 - 104	1997	1400	150	1130	85	API 5L X52	52000	N	1	CYL	78.00	5.50	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
12	CGP UNIT 15	Not Impacted	15-V-107	15-V-107	SUMPAL INLET SEPARATOR	GRK 15 - 99	1997	1250	125	1130	83	SA 516 Gr 70	20000	N	1	CYL	1575.00	57.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
13	CGP UNIT 15	Not Impacted	15-V-108	15-V-108	DAYUNG/SUMPAL PIPELINE PIG RECEIVER	GRK 15 - 99	2003	1400	150	1050	83	SA 516 Gr 70	20000	N	1	CYL	766.00	30.00	1.6	Wet H2S (HIC/SOHIC/Blistering) + Wet H2S (Sulfide Stress Cracking) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
14	CGP UNIT 15	Not Impacted	15-V-109	15-V-109	CONDENSATE PIG RECEIVER	GRK 15 - 52B	2006	1400	150	1120	150	Carbon Steel	20000	N	1	CYL	209.50	9.00	N/A	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)
15	CGP UNIT 15	Not Impacted	3K-KAQ-001	3K-KAQ-001	Tengah/Letang Rawa Pipeline Pig Receiver (CGPX)	GRK 15 - 155/156	2012	1350	150	1140	86	API 5L X52	52000	N	1	CYL	285.14	8.20	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
16	CGP UNIT 15	Not Impacted	TO-KAQ-001	TO-KAQ-001	SUBAN-GRISSIK TRUNKLINE PIG RECEIVER	GRK 15 - 155/156	2013	1440	200	74.46	98	API 5L X70	20000	N	1	CYL	733.45	14.27	0.9906	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
47	CGP UNIT 25	Not Impacted	24-HAL-101	24-HAL-101	TREATED GAS #1 COOLER	GRK 25 - 387	2012	1250	200	1119	130.6	SA 240 Gr 316L	16700	N	1	BOX	7188 (L) x 2686 (W) x 416 (H)	25.40	0	Chloride Stress Corrosion Cracking - External + Generic Failure Frequency	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)	LOW (2A)
55	CGP UNIT 25	Not Impacted	25-C-101	25-C-101 BOTTOM	AMINE CONTACTOR BOTTOM	GRK 25 - 317	1997	1250	200	1125	170	SA 516 Gr 70	20000	N	1	CYL	2199.89	88.00	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + Amine Corrosion + CO2 Corrosion + Low Temperature H2S Corrosion + Unspecified Int. Corr.	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
56	CGP UNIT 25	Not Impacted	25-C-101	25-C-101 TOP	AMINE CONTACTOR TOP	GRK 25 - 387	1997	1250	200	1119	170	SA 516 Gr 70	20000	N	1	CYL	2199.89	44.10	3.175	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM II (3C)		MEDIUM II (3C)		MEDIUM II (3C)	
84	CGP UNIT 25	Not Impacted	25-V-103	25-V-103	TREATED GAS SEPARATOR	GRK 25 - 384/386	1997	1250	150	1093	120	SA 516 Gr 70	20000	N	1	CYL	2210.00	88.00	0	Wet H2S (HIC/SOHIC/Blistering) + External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Low Temperature H2S Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
85	CGP UNIT 25	Not Impacted	25-V-105A	25-V-105A BOTTOM	TREATED GAS SEPARATOR	GRK 25-373A/373B	1997	195	400	150	323	SA 516 Gr 70	20000	Y	1	CYL	1219.20	12.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
110	CGP UNIT 30	Not Impacted	30-E-102	30-E-102 SHELL	GAS CHILLER & HC CODENSATE SUBCOOLER SHE	GRK 30-4515	1997	275	120	57	35	SA 516 Gr 70	17500	Y	1	CYL	1085.00	12.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
111	CGP UNIT 30	Not Impacted	30-E-103	30-E-103	PROPANE CONDENSER	GRK 30 - 414	1997	300	155	250	126	SA 516 Gr 70	20000	N	1	BOX	(248 (L) x 243 (W) x 152 (H))	16.25	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
112	CGP UNIT 30	Not Impacted	30-E-105	30-E-105	Propane Condenser	GRK 30 - 410	1997	275	185	250	120	SA 516 Gr 70	20000	N	0.85	BOX	(2543 (L) x 152 (W) x 243 (H))	16.00	3.175	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
113	CGP UNIT 30	Not Impacted	30-V-101	30-V-101	LOW TEMPERATURE SEPARATOR TOP	GRK 30-403	1997	1250	120	1096	52	SA 516 Gr 70	2000	Y	1	CYL	1981.20	78.00	0	CUI Carbon and Low Alloy Steels + CO2 Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)
114	CGP UNIT 30	Not Impacted	30-V-102	30-V-102	DEOPROPANIZER OVERHEADS DRUM	GRK 30 - 414	1997	300	150	175	120	SA 516 Gr 70	20000	N	1	CYL	590.94	9.53	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
115	CGP UNIT 30	Not Impacted	30-V-103	30-V-103	PROPANE STORAGE DRUM	GRK 30 - 414	1997	275	150	240	122	SA 516 Gr 70	20000	N	1	CYL	1676.40	18.00	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
116	CGP UNIT 30	Not Impacted	30-V-105	30-V-105	PROPANE COMPRESSOR SUCTION DRUM	GRK 30-409	1997	275	100	60	38	SA 516 Gr 70	20000	Y	1	CYL	1270.00	12.00	0	CUI Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
117	CGP UNIT 30	Not Impacted	30-V-106	30-V-106	REFRIGERANT BOILER	GRK 30 - 410	1997	275	150	175	120	SA 516 Gr 70	20000	N	1	CYL	336.55	8.00	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
118	CGP UNIT 30	Not Impacted	30-V-104	30-V-104	PROPANE ACCUMULATOR	GRK 30 - 407/411	1997	275	150	240	122	SA 516 Gr 70	20000	N	1	CYL	1676.40	18.00	0	External Corrosion Carbon and Low Alloy Steels + Unspecified Internal Corrosion + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
130	CGP UNIT 41	Not Impacted	41-C-101	41-C-101	GLYCOL CONTACTOR	GRK 41 - 316	1997	1250	175	1092	124	SA 516 Gr 70	20000	N	1	CYL	1350.00	55.00	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)
133	CGP UNIT 41	Not Impacted	41-S-101A	41-S-101 A	GLYCOL CATRIDGE FILTER	GRK 41 - 6-1	1997	150	200	75	171	SA 53B	15000	N	0.85	CYL	342.90	6.35	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)	MEDIUM II (3C)
134	CGP UNIT 41	Not Impacted	41-S-101B	41-S-101 B	GLYCOL CATRIDGE FILTER	GRK 41 - 6-1	1997	150	200	75	171	SA 53B	14600	N	0.85	CYL	342.90	6.35	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion + Chloride SCC Ext	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)	MEDIUM I (2C)
144	CGP UNIT 41	Not Impacted	41-V-103	41-V-103 HEAD	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-1	1997	150	200	75	175	SA 516 Gr 60	15000	N	0.85	CYL	1098.42	11.20	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	MEDIUM II (2D)	HIGH (3D)	HIGH (3D)
145	CGP UNIT 41	Not Impacted	41-V-103	41-V-103 SHELL	RICH GLYCOL FLASH SEPARATOR	GRK 41 - 4-1	1997	150	200	75	175	SA 516 Gr 60	15000	N	0.85	CYL	1098.42	9.90	3.175	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	MEDIUM II (2D)		MEDIUM II (2D)		MEDIUM II (2D)	
146	CGP UNIT 41	Not Impacted	41-V-104	41-V-104	GLYCOL ACCUMULATOR	GRK 41 - 8-1	1997	7	450	2	370	SA 516 Gr 60	20000	Y	0.85	CYL	1516.38	7.62	0	External Corrosion Carbon and Low Alloy Steels + CO2 Corrosion	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)	HIGH (3D)

APPENDIX C – REMAINING USEFUL LIFE ASSESSMENT RESULT

Static Equipment Directly Impacted by Post tie-in Condition

[illegible]

Static Equipment Indirectly Impacted by Post tie-in Condition

[illegible]

Static Equipment Not Impacted by Post tie-in Condition

APPENDIX D - LIFE EXTENSION PROGRAM BASED ON RISK AND EQUIPMENT CRITICALITY

Static Equipment Directly Impacted by Post tie-in Condition

Static Equipment Indirectly Impacted by Post tie-in Condition

ID	Name	Address	City	State	Country	Postal Code	Demographic Data				Economic Data				Social Data				Health Data				Education Data				Environment Data				Infrastructure Data				Safety Data				Quality of Life Data				Index Data																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
							Age Group	Gender	Ethnicity	Income Level	Unemployment Rate	Home Ownership	Median Income	Poverty Share	Life Expectancy	Healthcare Access	Education Level	High School Grad	College Grad	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend	Healthcare Access	Healthcare Spend																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
101	John Doe	123 Main St	New York	NY	USA	10001	25-34	M	White	High	5%	80%	\$75,000	10%	78	High	High School	College	\$1,200	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High	High

Static Equipment Not Impacted by Post tie-in Condition

ID	Name	Type	Status	Location	Description	Category	Material	Quantity	Unit	Value	Weight	Volume	Area	Length	Diameter	Thickness	Height	Width	Depth	Radius	Curvature	Taper	Flare	Bore	Hole	Slot	Notch	Groove	Ridge	Bulge	Dimple	Knurl	Rib	Flange	Bracket	Mount	Base	Foot	Leg	Arm	Shaft	Pin	Bolt	Nut	Washer	Gasket	Seal	O-ring	Valve	Fitting	Elbow	Teel	Coupling	Reducer	Adapter	Connector	Interface	Terminal	Plug	Cap	Cover	Door	Panel	Screen	Filter	Strainer	Sieve	Mesh	Screening	Filtration	Separation	Isolation	Control	Monitoring	Measurement	Detection	Alarm	Indicator	Signal	Data	Log	Record	Report	Summary	Analysis	Conclusion	Recommendation	Action	Status	Date	Time	User	Version	Revision	Change	History	Audit	Trace	Link	Path	URL	Access	Role	Permission	Privilege	Authority	Capability	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power	Energy	Force	Torque	Moment	Work	Heat	Power
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APPENDIX E – IAAN LIST

Date	Class	Asset Tag / Circuit ID	Damage	Description	System Status
12/19/2025	Static	25-C-201	Others	Flange bolt corroded	Closed
11/13/2023	Static	25-V-202	Others	Seepage on 2" condensate line Grissik	Closed
9/6/2022	Static	25-E-103	Internal corrosion	IAAN: INLET LINE OF COOLER 25-E-103C/D LEAK	Closed
7/11/2022	Static	25-E-102 A	External corrosion	Small Fugitive on Flange Connection	Closed
11/4/2019	Static	25-V-205 D	Others	LEAKAGE ON DS 25-LV-226 STEAM CONDENSATE	Closed
3/24/2013	Static	35-E-102	Others	Small Fugitive on Flange Connection	Closed
3/5/2012	Static	21-V-201 D	Internal Corrosion	Leakage on 10" outlet nozel(N2)_TSA201D	Closed
7/5/2011	Static	21-V-201 D	Others	leaking on 10"outlet nozzle (N2)_TSA201D	Closed
10/19/2008	Static	21-V-202	External Corrosion	Seepage on 2" condensate line grissik	Closed